chemotherapy vary with the regimen used but usually include myelosuppression, mucositis, nausea and vomiting, and nephrotoxicity (with cisplatin).

The mucosal side effects of therapy can lead to malnutrition and dehydration. Many centers address issues of dentition before starting treatment, and some place feeding tubes to ensure control of hydration and nutrition intake. About 50% of patients develop hypothyroidism from the treatment; thus, thyroid function should be monitored.

SALIVARY GLAND TUMORS

Most benign salivary gland tumors are treated with surgical excision, and patients with invasive salivary gland tumors are treated with surgery and radiation therapy. These tumors may recur regionally; adenoid cystic carcinoma has a tendency to recur along the nerve tracks. Distant metastases may occur as late as 10–20 years after the initial diagnosis. For metastatic disease, therapy is given with palliative intent, usually chemotherapy with doxorubicin and/or cisplatin. Identification of novel agents with activity in these tumors is a high priority. It is hoped that comprehensive genomic characterization of these rare tumors will facilitate these efforts.

FURTHER READING

Agrawal N et al: Exome sequencing of head and neck squamous cell carcinoma reveals inactivating mutations in NOTCH1. Science 333:1154, 2011.

Burtness B et al: Pembrolizumab alone or with chemotherapy versus cetuximab with chemotherapy for recurrent or metastatic squamous cell carcinoma of the head and neck (KEYNOTE-048): A randomized, open-label, phase 3 study. Lancet 394:1915, 2019.

Chow LQM: Head and neck cancer. N Engl J Med 382:60, 2020.

D'Cruz AK et al: Elective versus therapeutic neck dissection in node-negative oral cancer. N Engl J Med 373:521, 2015.

Ferris RL et al: Nivolumab for recurrent squamous-cell carcinoma of the head and neck. N Engl J Med 375:1856, 2016.

Gillison ML et al: Distinct risk factor profiles for human papillomavirus type 16-positive and human papillomavirus type 16-negative head and neck cancers. J Natl Cancer Inst 100:407, 2008.

Kang H et al: Whole-exome sequencing of salivary gland mucoepidermoid carcinoma. Clin Cancer Res 23:283, 2017.

Mehanna H et al: De-escalation after DE-ESCALATE and RTOG 1016: A Head and Neck Cancer Intergroup framework for future de-escalation studies. J Clin Oncol 38:2552, 2020.

Sabatini ME et al: Human papillomavirus as a driver of head and neck cancers. Br J Cancer 122:306, 2020.

Tota JE et al: Evolution of the oropharynx cancer epidemic in the United States: Moderation of increasing incidence in younger individuals and shift in the burden to older individuals. J Clin Oncol 37:1538, 2019.

78 Neoplasms of the Lung

Leora Horn, Wade T. Iams

Lung cancer, which was rare before 1900 with fewer than 400 cases described in the medical literature, is considered a disease of modern man, killing over three times as many men as prostate cancer and nearly twice as many women as breast cancer. Although lung cancer remains the number one cause of cancer-related mortality, a decline in lung cancer deaths has emerged, attributed to improvements in testing and therapeutic strategies and a decline in tobacco usage. Tobacco consumption is the primary cause of lung cancer, a reality firmly established in the mid-twentieth century and codified with the release of the U.S. Surgeon General's 1964 report on the health effects of tobacco smoking. Following the report, cigarette use started to decline in North America and parts of Europe, and with it, so did the incidence of lung cancer. Although tobacco smoking remains the primary cause of lung cancer worldwide, approximately 60% of new lung cancers in the United States occur in former smokers (smoked ≥100 cigarettes per lifetime, quit ≥1 year), many of whom quit decades ago, or never smokers (smoked <100 cigarettes per lifetime). Moreover, one in five women and one in 12 men diagnosed with lung cancer have never smoked.

EPIDEMIOLOGY

Lung cancer is the most common cause of cancer death among American men and women. Approximately 228,000 individuals will be diagnosed with lung cancer in the United States in 2020, and >135, 000 individuals will die from the disease. Lung cancer is uncommon below age 40, with rates increasing until age 80, after which the rate tapers off. The projected lifetime probability of developing lung cancer is estimated to be ~8% among males and ~6% among females. The incidence of lung cancer varies by racial and ethnic group, with the highest age-adjusted incidence rates among African Americans. The excess in age-adjusted rates among African Americans occurs only among men, but examinations of age-specific rates show that below age 50, mortality from lung cancer is >25% higher among African American than Caucasian women. Incidence and mortality rates among Hispanics and Native and Asian Americans are ~40–50% those of whites.

RISK FACTORS

Cigarette smokers have a 10-fold or greater increased risk of developing lung cancer compared to those who have never smoked. A large-scale genomic study suggested that one genetic mutation is induced for every 15 cigarettes smoked. The risk of lung cancer is lower among persons who quit smoking than among those who continue smoking. The size of the lung cancer risk reduction increases with the length of time the person has quit smoking, although even long-term former smokers have higher risks of lung cancer than those who never smoked. Cigarette smoking has been shown to increase the risk of all major types of lung cancer. Environmental tobacco smoke (ETS) or second-hand smoke is also an established cause of lung cancer. The risk from ETS is less than from active smoking, with about a 20–30% increase in lung cancer observed among never smokers married for many years to smokers, in comparison to the 2000% increase among continuing active smokers. The impact on the development of lung cancer among users of alternate nicotine delivery devices (e-cigarettes or vaping) is undefined. While one large randomized study demonstrated the superiority of e-cigarettes compared to traditional nicotine replacement therapy in aiding smoking cessation, e-cigarette– or vaping-associated lung injury (EVALI) is an emerging phenomenon that poses risks that may counterbalance the potential benefit in helping patients reduce traditional cigarette consumption and lung cancer risk.

Although cigarette smoking is the cause of the majority of lung cancers, several other risk factors have been identified, including occupational exposure to asbestos, arsenic, bischloromethyl ether, hexavalent chromium, mustard gas, nickel (as in certain nickel-refining processes), and polycyclic aromatic hydrocarbons.

Ionizing radiation is also an established lung carcinogen, most convincingly demonstrated from studies showing increased rates of lung cancer among survivors of the atom bombs dropped on Hiroshima and Nagasaki and large excesses among workers exposed to alpha irradiation from radon in underground uranium mining. Prolonged exposure to low-level radon in homes might impart a risk of lung cancer equal to or greater than that of ETS. Prior lung diseases such as chronic bronchitis, emphysema, and tuberculosis have been linked to increased risks of lung cancer as well. The risk of lung cancer appears to be higher among individuals with low fruit and vegetable intake during adulthood. This observation led to hypotheses that specific nutrients, in particular retinoids and carotenoids, might have chemopreventative effects for lung cancer. However, randomized trials failed to validate this hypothesis.

Smoking Cessation Given the undeniable link between cigarette smoking and lung cancer, physicians must promote tobacco abstinence. Stopping tobacco use before middle age avoids >90% of the lung cancer risk attributable to tobacco. Importantly, smoking cessation can even be beneficial in individuals with an established diagnosis of lung cancer, as it is associated with improved survival, fewer side effects from therapy, and an overall improvement in quality of life. Consequently, it is important to promote smoking cessation even *after* the diagnosis of lung cancer is established.

Physicians need to understand the essential elements of smoking cessation therapy. The individual must want to stop smoking and must be willing to work hard to achieve the goal of smoking abstinence. Self-help strategies alone only marginally affect quit rates, whereas individual and combined pharmacotherapies in combination with counseling can significantly increase rates of cessation. Therapy with an antidepressant (e.g., bupropion) and nicotine replacement therapy (varenicline, a $\alpha_4\beta_2$ nicotinic acetylcholine receptor partial agonist) are approved by the U.S. Food and Drug Administration (FDA) as first-line treatments for nicotine dependence. In a randomized trial, varenicline was shown to be more efficacious than bupropion or placebo. Prolonged use of varenicline beyond the initial induction phase proved useful in maintaining smoking abstinence. Clonidine and nortriptyline are recommended as second-line treatments. A role for e-cigarettes has not been definitively established **(Chap. 454)**.

Inherited Predisposition to Lung Cancer Exposure to environmental carcinogens, such as those found in tobacco smoke, induce or facilitate the transformation from bronchoepithelial cells to a malignant phenotype. The contribution of carcinogens to transformation is modulated by polymorphic variations in genes that affect aspects of carcinogen metabolism. Certain genetic polymorphisms of the P450 enzyme system, specifically CYP1A1, and chromosome fragility are associated with the development of lung cancer. These genetic variations occur at relatively high frequency in the population, but their contribution to an individual's lung cancer risk is generally low. However, because of their population frequency, the overall impact on lung cancer risk could be high.

First-degree relatives of lung cancer probands have a two- to three-fold excess risk of lung cancer and other cancers, many of which are not smoking-related. These data suggest that specific genes and/or genetic variants may contribute to susceptibility to lung cancer. However, very few such genes have yet been identified. Individuals with inherited mutations in *RB* (patients with retinoblastoma living to adulthood) and *p53* (Li-Fraumeni syndrome) genes may develop lung cancer. Common gene variants involved in lung cancer have identified three separate loci that are associated with lung cancer (5p15, 6p21, and 15q25) and include genes that regulate acetylcholine nicotinic receptors and telomerase production. A rare germline mutation (T790M) involving the epidermal growth factor receptor (EGFR) maybe be linked to lung cancer susceptibility in never smokers. Likewise, a susceptibility locus on chromosome 6q greatly increases lung cancer risk among light and never smokers. Although progress has been made, there is a significant amount of work that remains to be done in identifying heritable risk factors for lung cancer. Currently no molecular criteria are suitable to select patients for more intense screening programs or for specific chemopreventive strategies.

PATHOLOGY

The World Health Organization (WHO) defines lung cancer as tumors arising from the respiratory epithelium (bronchi, bronchioles, and alveoli). The WHO classification system divides epithelial lung cancers into four major cell types: small-cell lung cancer (SCLC), adenocarcinoma, squamous cell carcinoma, and large-cell carcinoma; the latter three types are collectively known as non-small-cell carcinomas (NSCLCs) **(Fig. 78-1)**. Small-cell carcinomas consist of small cells with scant cytoplasm, ill-defined cell borders, finely granular nuclear chromatin, absent or inconspicuous nucleoli, and a high mitotic count. SCLC may be distinguished from NSCLC by the presence of neuroendocrine markers including CD56, neural cell adhesion molecule (NCAM), synaptophysin, and chromogranin. Adenocarcinomas possess glandular differentiation or mucin production and may show acinar, papillary, lepidic, or solid features or a mixture of these patterns. Squamous cell carcinomas of the lung are morphologically identical to extrapulmonary squamous cell carcinomas and cannot be distinguished by immunohistochemistry alone. Squamous cell tumors show keratinization and/or intercellular bridges that arise from bronchial epithelium. The tumor consists of sheets of cells rather than the three-dimensional groups of cells characteristic of adenocarcinomas. Large-cell carcinomas compose <10% of lung carcinomas. These tumors lack the cytologic and architectural features of small-cell carcinoma and glandular or squamous differentiation. Together, these four histologic types account for ~90% of all epithelial lung cancers.

All histologic types of lung cancer can develop in current and former smokers, although squamous and small-cell carcinomas are most commonly associated with tobacco use. With the decline in cigarette consumption, adenocarcinoma has become the most frequent histologic subtype of lung cancer in the United States. In lifetime never smokers or former light smokers (<10 pack-year history), women, and younger adults (<60 years), adenocarcinoma tends to be the most common form of lung cancer.

In addition to distinguishing between SCLC and NSCLC, because these tumors have quite different natural histories and therapeutic approaches (see below), it is necessary to classify whether NSCLC is squamous or nonsquamous. The classification system, developed jointly by the International Association for the Study of Lung Cancer, the American Thoracic Society, and the European Respiratory Society, provides an integrated approach to the classification of lung adenocarcinoma that includes clinical, molecular, radiographic, and pathologic information.

It is recognized that most lung cancers present in an advanced stage and are often diagnosed based on small biopsies or cytologic specimens, rendering clear histologic distinctions difficult, if not impossible. In such cases, particularly in patients with advanced-stage disease, a repeat biopsy is recommended to obtain additional tissue for further clarification. The distinction between squamous and nonsquamous lung cancer is viewed as critical to optimal therapeutic decision making, and a diagnosis of *non-small-cell carcinoma, not otherwise specified* is no longer considered acceptable. This distinction can be achieved using a single marker for adenocarcinoma (thyroid transcription factor-1 or napsin-A) plus a squamous marker (p40 or p63) and/or mucin stains. If tissue is limited and a clear morphologic pattern is evident, a diagnosis can be made without immunohistochemistry staining. In addition to determining histologic subtype, preservation of sufficient specimen material for appropriate molecular testing and PD-L1 testing necessary to help guide therapeutic decision making is recommended (see below).

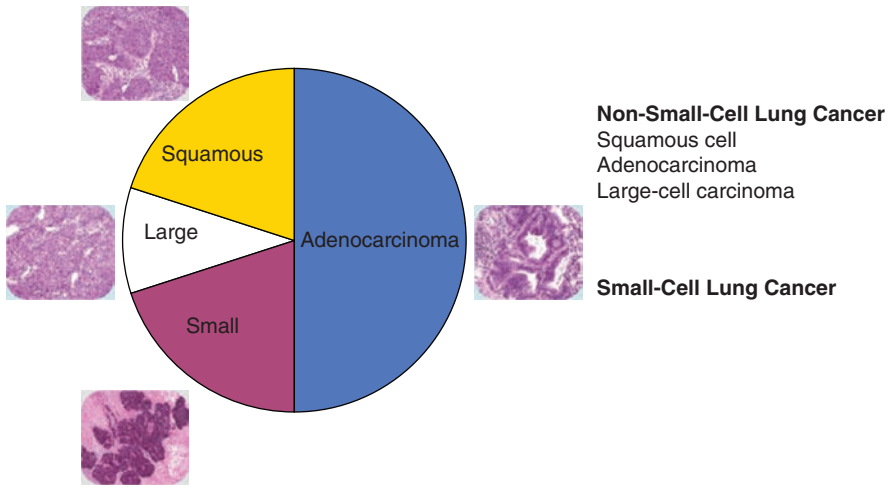


FIGURE 78-1 Histologic subsets of lung cancer.

The terms *adenocarcinoma in situ* and *minimally invasive adenocarcinoma* are now recommended for small solitary adenocarcinomas (≤3 cm) with either pure lepidic growth (term used to describe single-layered growth of atypical cuboidal cells coating the alveolar walls) or predominant lepidic growth with ≤5 mm invasion. Individuals with these entities experience 100% or near 100% 5-year disease-free survival with complete tumor resection. *Invasive adenocarcinomas*, representing more than 70–90% of surgically resected lung adenocarcinomas, are now classified by their predominant pattern: lepidic, acinar, papillary, and solid patterns. Lepidic-predominant subtype has a favorable prognosis, acinar and papillary have an intermediate prognosis, and solid-predominant has a poor prognosis. The terms *signet ring* and *clear cell adenocarcinoma* have been eliminated from the variants of invasive lung adenocarcinoma, whereas the term *micropapillary*, a subtype with a particularly poor prognosis, has been added. Because of prognostic implications, squamous cell carcinoma has also been modified to consist of keratinizing, nonkeratinizing, and basaloid, analogous to head and neck cancers.

IMMUNOHISTOCHEMISTRY

The diagnosis of lung cancer most often rests on the morphologic or cytologic features correlated with clinical and radiographic findings. Immunohistochemistry may be used to verify neuroendocrine differentiation within a tumor, with markers such as neuron-specific enolase (NSE), CD56 or NCAM, synaptophysin, chromogranin, and Leu7. Immunohistochemistry is also helpful in differentiating primary from metastatic adenocarcinomas; thyroid transcription factor-1 (TTF-1), identified in tumors of thyroid and pulmonary origin, is positive in >70% of pulmonary adenocarcinomas and is a reliable indicator of primary lung cancer, provided a thyroid primary has been excluded. A negative TTF-1, however, does not exclude the possibility of a lung primary. TTF-1 is also positive in neuroendocrine tumors of pulmonary and extrapulmonary origin. Napsin-A (Nap-A) is an aspartic protease that plays an important role in maturation of surfactant B7 and is expressed in cytoplasm of type II pneumocytes. In several studies, Nap-A has been reported in >90% of primary lung adenocarcinomas. Notably, a combination of Nap-A and TTF-1 is useful in distinguishing primary lung adenocarcinoma (Nap-A positive, TTF-1 positive) from primary lung squamous cell carcinoma (Nap-A negative, TTF-1 negative) and primary SCLC (Nap-A negative, TTF-1 positive). Cytokeratins 7 and 20 used in combination can help narrow the differential diagnosis; nonsquamous NSCLC, SCLC, and mesothelioma may stain positive for CK7 and negative for CK20, whereas squamous cell lung cancer often will be both CK7 and CK20 negative. p63 is a useful marker for the detection of NSCLCs with squamous differentiation when used in cytologic pulmonary samples. Mesothelioma can be easily identified ultrastructurally, but it has historically been difficult to differentiate from adenocarcinoma through morphology and immunohistochemical staining. Several markers in the past few years have proven to be more helpful including CK5/6, calretinin, and Wilms tumor gene-1 (*WT-1*), all of which show positivity in mesothelioma.

MOLECULAR PATHOGENESIS

As proposed by Hanahan and Weinberg, virtually all cancer cells acquire six hallmark capabilities: self-sufficiency in growth signals, insensitivity to antigrowth signals, evading apoptosis, limitless replicative potential, sustained angiogenesis, and tissue invasion and metastasis. The order in which these hallmark capabilities are acquired is variable. Events leading to acquisition of these hallmarks vary widely, although broadly, cancers arise as a result of accumulations of gain-of-function mutations in oncogenes and loss-of-function mutations in tumor-suppressor genes. Further complicating the study of lung cancer, the sequence of events that leads to disease is clearly different for the various histopathologic entities.

For cancers in general, one theory holds that a small subset of the cells within a tumor (i.e., "stem cells") are responsible for the full malignant behavior of the tumor. As part of this concept, the large bulk of the cells in a cancer are "offspring" of these cancer stem cells. While clonally related to the cancer stem cell subpopulation, most cells by themselves cannot regenerate the full malignant phenotype. The stem cell concept may explain the failure of standard medical therapies to eradicate lung cancers, even when there is a clinical complete response. Disease recurs because therapies do not eliminate the stem cell component, which may be more resistant to therapy. Precise human lung cancer stem cells have yet to be identified.

Among lung cancer histologies, adenocarcinomas have been the most extensively catalogued for recurrent genomic gains and losses as well as for somatic mutations **(Fig. 78-2, Table 78-1)**. While multiple different kinds of aberrations have been found, a major class involves "driver mutations," which are mutations that occur in genes encoding signaling proteins that, when aberrant, drive initiation and maintenance of tumor cells. Importantly, driver mutations can serve as a potential Achilles' heels for tumors, if their gene products can be targeted appropriately. These genes encode cell-surface receptors consisting of an extracellular ligand-binding domain, a transmembrane structure, and an intracellular tyrosine kinase (TK) domain. The binding of ligand to receptor activates receptor dimerization and TK autophosphorylation, initiating a cascade of intracellular events, and leading to increased cell proliferation, angiogenesis, metastasis, and a decrease in apoptosis. Lung adenocarcinomas can arise when normal alveolar type II cells develop mutations in *EGFR*, *BRAF*, *MET*, *KRAS*, and *PIK3CA*. These same tumors display high sensitivity to small-molecule TK inhibitors (TKIs). Additional subsets of lung adenocarcinoma have been identified as defined by the presence of specific chromosomal rearrangements, resulting in the aberrant activation of the TKs *ALK*, *ROS1*, *NTRK*, and *RET*. Notably, most driver mutations in lung cancer appear to be mutually exclusive, suggesting that acquisition of one of these mutations is sufficient to drive tumorigenesis. Although driver mutations have mostly been found in adenocarcinomas, three

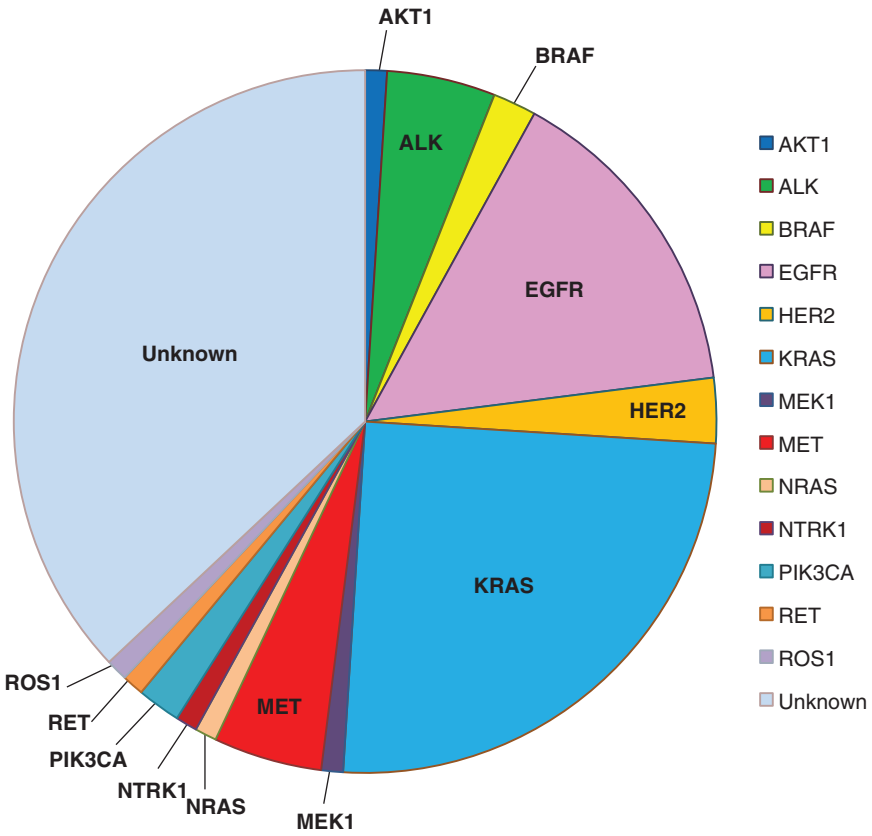


FIGURE 78-2 Driver mutations in lung adenocarcinomas.

TABLE 78-1 Driver Mutations in Non-Small-Cell Lung Cancer (NSCLC)

GENE	ALTERATION	FREQUENCY IN NSCLC	TYPICAL HISTOLOGY
AKT1	Mutation	1%	Adenocarcinoma, squamous
ALK	Rearrangement	3–7%	Adenocarcinoma
BRAF	Mutation	1–3%	Adenocarcinoma
DDR2	Mutation	~4%	Squamous
EGFR	Mutation	10–35%	Adenocarcinoma
FGFR1	Amplification	~20%	Squamous
HER2	Mutation	2–4%	Adenocarcinoma
KRAS	Mutation	15–25%	Adenocarcinoma
MEK1	Mutation	1%	Adenocarcinoma
MET	Amplification	2–4%	Adenocarcinoma
NRAS	Mutation	1%	Adenocarcinoma
NTRK	Rearrangement	1–2%	Adenocarcinoma
PIK3CA	Mutation	1–3%	Squamous
PTEN	Mutation	4–8%	Squamous
ROS1	Rearrangement	1–2%	Adenocarcinoma

potential molecular targets have been identified in squamous cell lung carcinomas: *FGFR1* amplification, *DDR2* mutations, and *PIK3CA* mutations/*PTEN* loss as well as *BRAF* and *MET* (Table 78-1).

A large number of tumor-suppressor genes have also been identified that are inactivated during the pathogenesis of lung cancer. These include *TP53*, *RB1*, *RASSF1A*, *CDKN2A/B*, *LKB1* (*STK11*), and *FHIT*. Nearly 90% of SCLCs harbor mutations in *TP53* and *RB1*. Several tumor-suppressor genes on chromosome 3p appear to be involved in nearly all lung cancers. Allelic loss of this region occurs very early in lung cancer pathogenesis, including in histologically normal smoking-damaged lung epithelium.

EARLY DETECTION AND SCREENING

In lung cancer, clinical outcome is related to the stage at diagnosis, and hence, it is generally assumed that early detection of occult tumors will lead to improved survival. Early detection is a process that involves screening tests, surveillance, diagnosis, and early treatment. Screening refers to the use of tests across a healthy population in order to identify individuals who harbor asymptomatic disease. For a screening program to be successful, the target population must have a high burden of disease; the test must be sensitive, specific, accessible, and cost effective; and effective treatment must be available that can reduce mortality. With any screening procedure, it is important to consider the possible influence of *lead-time bias* (detecting the cancer earlier without an effect on survival), *length-time bias* (indolent cancers are detected on screening and may not affect survival, whereas aggressive cancers are likely to cause symptoms earlier in patients and are less likely to be detected), and *overdiagnosis* (diagnosing cancers so slow growing that they are unlikely to cause the death of the patient).

Because a majority of lung cancer patients present with advanced disease beyond the scope of surgical resection, the value of screening for this condition is debated. Indeed, randomized controlled trials conducted in the 1960s to 1980s using screening chest x-rays (CXR), with or without sputum cytology, reported no impact on lung cancer–specific mortality in patients characterized as high risk (males age ≥45 years with a smoking history). These studies have been criticized for their design, statistical analyses, and outdated imaging modalities. In contrast to CXR, low-dose, noncontrast, thin-slice spiral chest computed tomography (LDCT) has emerged as an effective tool to screen for lung cancer. In nonrandomized studies conducted in the 1990s, LDCT scans were shown to detect more lung nodules and cancers than standard CXR in selected high-risk populations (e.g., age ≥60 years and a smoking history of ≥10 pack-years). Notably, up to 85% of the lung cancers discovered in these trials were classified as stage I disease and therefore considered potentially curable with surgical resection.

These data prompted the National Cancer Institute (NCI) to initiate the National Lung Screening Trial (NLST), a randomized study designed to determine if LDCT screening could reduce mortality from lung cancer in high-risk populations as compared with standard posterior anterior CXR. High-risk patients were defined as individuals between 55 and 74 years of age, with a ≥30 pack-year history of cigarette smoking; former smokers must have quit within the previous 15 years. Excluded from the trial were individuals with a previous lung cancer diagnosis, a history of hemoptysis, an unexplained weight loss of >15 lb in the preceding year, or a chest CT within 18 months of enrollment. A total of 53,454 persons were enrolled and randomized to annual screening yearly for 3 years (LDCT screening, n = 26,722; CXR screening, n = 26,732). Any noncalcified nodule measuring ≥4 mm in any diameter found on LDCT and CXR images with any noncalcified nodule or mass were classified as "positive." Participating radiologists had the option of not calling a final screen positive if a noncalcified nodule had been stable on the three screening examinations. Overall, 39.1% of participants in the LDCT group and 16% in the CXR group had at least one positive screening result. Of those who screened positive, the false-positive rate was 96.4% in the LDCT group and 94.5% in the CXR group. This was consistent across all three rounds. In the LDCT group, 1060 cancers were identified compared with 941 cancers in the CXR group (645 vs 572 per 100,000 person-years; relative risk [RR], 1.13; 95% confidence interval [CI], 1.03–1.23). Nearly twice as many stage IA cancers were detected in the LDCT group compared with the CXR group (40% vs 21%). The overall rates of lung cancer death were 247 and 309 deaths per 100,000 participants in the LDCT and CXR groups, respectively, representing a 20% reduction in lung cancer mortality in the LDCT-screened population (95% CI, 6.8–26.7%; $p = .004$). Compared with the CXR group, the rate of death in the LDCT group from *any* cause was reduced by 6.7% (95% CI, 1.2–13.6%; $p = .02$). The number needed to screen (NNTS) to prevent one lung cancer death was calculated to be 320.

The Nelson study was a second randomized trial comparing no screening to CT scans at baseline and in years 1, 3, and 5.5 in 13,195 men and 2594 women. Participants were 50–75 years of age and were current and former smokers with 10 years or less of cessation who smoked >15 cigarettes a day for >25 years or >10 cigarettes daily for >30 years. Participants were selected from four regions in the Netherlands or Belgium and were excluded if they were in moderate or bad self-reported health, were unable to climb two flights of stairs, had a body weight >140 kg, had a CT of the chest within the past year or a history of lung cancer <5 years ago or were still under treatment, or had current or past renal cell carcinoma, melanoma, or breast cancer. The hazard ratio for lung cancer mortality at 10 years was 0.74 (95% CI, 0.60–0.91; $p = .003$) and 0.61 (95% CI, 0.35–1.04; $p = .0543$) in men and women, respectively. These two trials have validated the use of annual CT scans for early detection of lung cancer in high-risk populations.

LDCT screening for lung cancer comes with known risks including a high rate of false-positive results, false-negative results, potential for unnecessary follow-up testing, radiation exposure, overdiagnosis, changes in anxiety level and quality of life, and substantial financial costs. By far, the biggest challenge confronting the use of CT screening is the high false-positive rate. False positives can have a substantial impact on patients through the expense and risk of unneeded further evaluation and emotional stress. The management of these patients usually consists of serial CT scans over time to see if the nodules grow, attempted fine-needle aspirates, or surgical resection. At $300 per scan (NCI estimated cost), the outlay for initial LDCT alone could run into the billions of dollars annually, an expense that only further escalates when factoring in various downstream expenditures an individual might incur in the assessment of positive findings. A formal cost-effectiveness analysis of the NLST demonstrated differences between sex, age, and current smoking status and the method of follow-up. Despite some questions, LDCT screening has been recommended for all patients meeting criteria for enrollment on NLST. When discussing the option of LDCT screening, use of absolute risks rather than relative risks is helpful because studies indicate the public can process absolute terminology more effectively than relative risk projections. A useful

TABLE 78-2 The Benefits and Harms of LDCT Screening for Lung Cancer Based on NLST Data

	LDCT	CXR
Benefits: How did CT scans help compared to CXR?		
4 in 1000 fewer died from lung cancer	13 in 1000	17 in 1000
5 in 1000 fewer died from all causes	70 in 1000	75 in 1000
Harms: What problems did CT scans cause compared to CXR?		
223 in 1000 had at least 1 false alarm	365 in 1000	142 in 1000
18 in 1000 had a false alarm leading to an invasive procedure	25 in 1000	7 in 1000
2 in 1000 had a major complication from an invasive procedure	3 in 1000	1 in 1000

Abbreviations: CXR, chest x-ray; LDCT, low-dose computed tomography; NLST, National Lung Screening Trial.

Source: From S Woloshin: Cancer screening campaigns getting past uninformative persuasion. N Engl J Med 367:1167, 2012. Copyright © (2012) Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

guide has been developed by the NCI to help patients and physicians assess the benefits and harms of LDCT screening for lung cancer (Table 78-2).

CLINICAL MANIFESTATIONS

Over half of all patients diagnosed with lung cancer present with locally advanced or metastatic disease at the time of diagnosis. The majority of patients present with signs, symptoms, or laboratory abnormalities that can be attributed to the primary lesion, local tumor growth, invasion or obstruction of adjacent structures, growth at distant metastatic sites, or a paraneoplastic syndrome (Tables 78-3 and 78-4). The prototypical lung cancer patient is a current or former smoker of either sex, usually in the seventh decade of life. A history of chronic cough with or without hemoptysis in a current or former smoker with chronic obstructive pulmonary disease (COPD) age 40 years or older should prompt a thorough investigation for lung cancer even in the face of a normal CXR. A persistent pneumonia without constitutional symptoms and unresponsive to repeated courses of antibiotics also should prompt an evaluation for the underlying cause. Lung cancer arising in a lifetime never smoker is more common in women and East Asians. Such patients also tend to be younger than their smoking counterparts at the time of diagnosis. The clinical presentation of lung cancer in never smokers tends to mirror that of current and former smokers.

Patients with central or endobronchial growth of the primary tumor may present with cough, hemoptysis, wheeze, stridor, dyspnea, or

TABLE 78-3 Presenting Signs and Symptoms of Lung Cancer

SYMPTOM AND SIGNS	RANGE OF FREQUENCY
Cough	8–75%
Weight loss	0–68%
Dyspnea	3–60%
Chest pain	20–49%
Hemoptysis	6–35%
Bone pain	6–25%
Clubbing	0–20%
Fever	0–20%
Weakness	0–10%
Superior vena cava obstruction	0–4%
Dysphagia	0–2%
Wheezing and stridor	0–2%

Source: Reproduced with permission from MA Beckles: Initial evaluation of the patient with lung cancer. Symptoms, sign, laboratory tests, and paraneoplastic syndromes. Chest 123:97, 2003.

TABLE 78-4 Clinical Findings Suggestive of Metastatic Disease

Symptoms elicited in history	• Constitutional: weight loss >10 lb • Musculoskeletal: pain • Neurologic: headaches, syncope, seizures, extremity weakness, recent change in mental status
Signs found on physical examination	• Lymphadenopathy (>1 cm) • Hoarseness, superior vena cava syndrome • Bone tenderness • Hepatomegaly (>13 cm span) • Focal neurologic signs, papilledema • Soft-tissue mass
Routine laboratory tests	• Hematocrit, <40% in men; <35% in women • Elevated alkaline phosphatase, GGT, SGOT, and calcium levels

Abbreviations: GGT, gamma-glutamyltransferase; SGOT, serum glutamic-oxaloacetic transaminase.

Source: Reproduced with permission from GA Silvestri et al: The noninvasive staging of non-small cell lung cancer. Chest 123:147S, 2003.

postobstructive pneumonia. Peripheral growth of the primary tumor may cause pain from pleural or chest wall involvement, dyspnea on a restrictive basis, and symptoms of a lung abscess resulting from tumor cavitation. Regional spread of tumor in the thorax (by contiguous growth or by metastasis to regional lymph nodes) may cause tracheal obstruction, esophageal compression with dysphagia, recurrent laryngeal nerve paralysis with hoarseness, phrenic nerve palsy with elevation of the hemidiaphragm and dyspnea, and sympathetic nerve paralysis with Horner's syndrome (enophthalmos, ptosis, miosis, and anhidrosis). Malignant pleural effusions can cause pain, dyspnea, or cough. Pancoast (or superior sulcus tumor) syndromes result from local extension of a tumor growing in the apex of the lung with involvement of the eighth cervical and first and second thoracic nerves, and present with shoulder pain that characteristically radiates in the ulnar distribution of the arm, often with radiologic destruction of the first and second ribs. Often Horner's syndrome and Pancoast syndrome coexist. Other problems of regional spread include superior vena cava syndrome from vascular obstruction; pericardial and cardiac extension with resultant tamponade, arrhythmia, or cardiac failure; lymphatic obstruction with resultant pleural effusion; and lymphangitic spread through the lungs with hypoxemia and dyspnea. In addition, lung cancer can spread transbronchially, producing tumor growth along multiple alveolar surfaces with impairment of gas exchange, respiratory insufficiency, dyspnea, hypoxemia, and sputum production. Constitutional symptoms may include anorexia, weight loss, weakness, fever, and night sweats. Apart from the brevity of symptom duration, these parameters fail to clearly distinguish SCLC from NSCLC or even from neoplasms metastatic to lungs.

Extrathoracic metastatic disease is found at autopsy in >50% of patients with squamous carcinoma, 80% of patients with adenocarcinoma and large-cell carcinoma, and >95% of patients with SCLC. Approximately one-third of patients present with symptoms as a result of distant metastases. Lung cancer metastases may occur in virtually every organ system, and the site of metastatic involvement largely determines other symptoms. Patients with brain metastases may present with headache, nausea and vomiting, seizures, or neurologic deficits. Patients with bone metastases may present with pain, pathologic fractures, or spinal cord compression. The latter may also occur with epidural metastases. Individuals with bone marrow invasion may present with cytopenias or leukoerythroblastosis. Those with liver metastases may present with hepatomegaly, right upper quadrant pain, fever, anorexia, and weight loss. Liver dysfunction and biliary obstruction are rare. Adrenal metastases are common but rarely cause pain or adrenal insufficiency unless they are large.

Paraneoplastic syndromes are common in patients with lung cancer, especially those with SCLC, and may be the presenting finding or the first sign of recurrence. In addition, paraneoplastic syndromes may mimic metastatic disease and, unless detected, lead to inappropriate

palliative rather than curative treatment. Often the paraneoplastic syndrome may be relieved with successful treatment of the tumor. In some cases, the pathophysiology of the paraneoplastic syndrome is known, particularly when a hormone with biologic activity is secreted by a tumor. However, in many cases, the pathophysiology is unknown. Systemic symptoms of anorexia, cachexia, weight loss (seen in 30% of patients), fever, and suppressed immunity are paraneoplastic syndromes of unknown etiology or at least not well defined. Weight loss >10% of total body weight is considered a bad prognostic sign. Endocrine syndromes are seen in 12% of patients; hypercalcemia resulting from ectopic production of parathyroid hormone (PTH) or, more commonly, PTH-related peptide is the most common life-threatening metabolic complication of malignancy, primarily occurring with squamous cell carcinomas of the lung. Clinical symptoms include nausea, vomiting, abdominal pain, constipation, polyuria, thirst, and altered mental status.

Hyponatremia may be caused by the syndrome of inappropriate secretion of antidiuretic hormone (SIADH) or possibly atrial natriuretic peptide (ANP) **(Chap. 93)**. SIADH resolves within 1–4 weeks of initiating chemotherapy in the vast majority of cases. During this period, serum sodium can usually be managed and maintained above 128 mEq/L via fluid restriction. Demeclocycline can be a useful adjunctive measure when fluid restriction alone is insufficient. Vasopressin receptor antagonists like tolvaptan also have been used in the management of SIADH. However, the use of tolvaptan has significant limitations including liver injury and overly rapid correction of the hyponatremia, which can lead to irreversible neurologic injury. Likewise, the cost of tolvaptan may be prohibitive (as high as $300 per tablet in some areas). Of note, patients with ectopic ANP may have worsening hyponatremia if sodium intake is not concomitantly increased. Accordingly, if hyponatremia fails to improve or worsens after 3–4 days of adequate fluid restriction, plasma levels of ANP should be measured to determine the causative syndrome.

Ectopic secretion of ACTH by SCLC and pulmonary carcinoids usually results in additional electrolyte disturbances, especially hypokalemia, rather than the changes in body habitus that occur in Cushing's syndrome from a pituitary adenoma **(Chap. 93)**. Treatment with standard medications, such as metyrapone and ketoconazole, is largely ineffective due to extremely high cortisol levels. The most effective strategy for management of the Cushing's syndrome is effective treatment of the underlying SCLC. Bilateral adrenalectomy may be considered in extreme cases.

Skeletal–connective tissue syndromes include clubbing in 30% of cases (usually NSCLCs) and hypertrophic primary osteoarthropathy in 1–10% of cases (usually adenocarcinomas). Patients may develop periostitis, causing pain, tenderness, and swelling over the affected bones and a positive bone scan. Neurologic–myopathic syndromes are seen in only 1% of patients but are dramatic and include the myasthenic Eaton-Lambert syndrome and retinal blindness with SCLC, whereas peripheral neuropathies, subacute cerebellar degeneration, cortical degeneration, and polymyositis are seen with all lung cancer types. Many of these are caused by autoimmune responses such as the development of anti-voltage-gated calcium channel antibodies in Eaton-Lambert syndrome. Patients with this disorder present with proximal muscle weakness, usually in the lower extremities, occasional autonomic dysfunction, and rarely, cranial nerve symptoms or involvement of the bulbar or respiratory muscles. Depressed deep tendon reflexes are frequently present. In contrast to patients with myasthenia gravis, strength improves with serial effort. Some patients who respond to chemotherapy will have resolution of the neurologic abnormalities. Thus, chemotherapy is the initial treatment of choice. Paraneoplastic encephalomyelitis and sensory neuropathies, cerebellar degeneration, limbic encephalitis, and brainstem encephalitis occur in SCLC in association with a variety of antineuronal antibodies such as anti-Hu, anti-CRMP5, and ANNA-3. Paraneoplastic cerebellar degeneration may be associated with anti-Hu, anti-Yo, or P/Q calcium channel autoantibodies. Coagulation or thrombotic or other hematologic manifestations occur in 1–8% of patients and include migratory venous thrombophlebitis (Trousseau's syndrome), nonbacterial thrombotic (marantic) endocarditis with arterial emboli, and disseminated intravascular coagulation with hemorrhage, anemia, granulocytosis, and leukoerythroblastosis. Thrombotic disease complicating cancer is usually a poor prognostic sign. Cutaneous manifestations such as dermatomyositis and acanthosis nigricans are uncommon (1%), as are the renal manifestations of nephrotic syndrome and glomerulonephritis (≤1%).

DIAGNOSING LUNG CANCER

Tissue sampling is required to confirm a diagnosis in all patients with suspected lung cancer. In patients with suspected metastatic disease, a biopsy of a distant site of disease is preferred for tissue confirmation. Given the greater emphasis placed on molecular and PD-L1 testing for NSCLC patients, a core biopsy is preferred to ensure adequate tissue for analysis. Tumor tissue may be obtained via minimally invasive techniques such as bronchial or transbronchial biopsy during fiberoptic bronchoscopy, by fine-needle aspiration (FNA) or percutaneous biopsy using image guidance, or via endobronchial ultrasound (EBUS)-guided biopsy. Depending on the location, lymph node sampling may occur via transesophageal endoscopic ultrasound (EUS)-guided biopsy, EBUS-guided biopsy, or blind biopsy. In patients with suspected metastatic disease, a diagnosis may be confirmed by bronchoscopy, percutaneous biopsy of a soft tissue mass, lytic bone lesion, bone marrow, pleural or liver lesion, or an adequate cell block obtained from a malignant pleural effusion. In patients with a suspected malignant pleural effusion, if the initial thoracentesis is negative, a repeat thoracentesis is warranted. Although the majority of pleural effusions are due to malignant disease, particularly if they are exudative or bloody, some may be parapneumonic. In the absence of distant disease, such patients should be considered for possible curative treatment.

The diagnostic yield of any biopsy depends on several factors including location (accessibility) of the tumor, tumor size, tumor type, and technical aspects of the diagnostic procedure including the experience level of the bronchoscopist and pathologist. In general, central lesions such as squamous cell carcinomas, small-cell carcinomas, or endobronchial lesions such as carcinoid tumors are more readily diagnosed by bronchoscopic examination, whereas peripheral lesions such as adenocarcinomas and large-cell carcinomas are more amenable to transthoracic biopsy.

Sputum cytology is inexpensive and noninvasive but has a lower yield than other specimen types due to poor preservation of the cells and more variability in acquiring a good-quality specimen. The yield for sputum cytology is highest for larger and centrally located tumors such as squamous cell carcinoma and small-cell carcinoma histology. The specificity for sputum cytology averages close to 100%, although sensitivity is generally <70%. The accuracy of sputum cytology improves with increased numbers of specimens analyzed. Consequently, analysis of at least three sputum specimens is recommended. However, the quality of the specimen may not be adequate for histologic subclassification and PD-L1 and molecular testing.

STAGING LUNG CANCER

Lung cancer staging consists of two parts: first, a determination of the location of the tumor and possible metastatic sites (anatomic staging), and second, an assessment of a patient's ability to withstand various antitumor treatments (physiologic staging). All patients with lung cancer should have a complete history and physical examination, with evaluation of all other medical problems, determination of performance status, and history of weight loss. Staging with regard to a patient's potential for surgical resection is principally applicable to NSCLC.

ANATOMIC STAGING OF PATIENTS WITH LUNG CANCER

The accurate staging of patients with NSCLC is essential for determining the appropriate treatment in patients with resectable disease and for avoiding unnecessary surgical procedures in patients with advanced disease. All patients with NSCLC should undergo initial radiographic imaging with CT scan, positron emission tomography (PET), or preferably CT-PET. PET scanning attempts to identify sites of malignancy based on glucose metabolism by measuring the uptake

of ^{18}F-fluorodeoxyglucose (FDG). Rapidly dividing cells, presumably in the lung tumors, will preferentially take up ^{18}F-FDG and appear as a "hot spot." To date, PET has been mostly used for staging and detection of metastases in lung cancer and in the detection of nodules >15 mm in diameter. Combined ^{18}F-FDG PET-CT imaging has been shown to improve the accuracy of staging in NSCLC compared to visual correlation of PET and CT or either study alone. CT-PET has been found to be superior in identifying pathologically enlarged mediastinal lymph nodes and extrathoracic metastases. A standardized uptake value (SUV) of >2.5 on PET is highly suspicious for malignancy. False negatives can be seen in diabetes, in lesions <8 mm, and in slow-growing tumors (e.g., carcinoid tumors or well-differentiated adenocarcinoma). False positives can be seen in certain infections and granulomatous disease (e.g., tuberculosis). Thus, PET should never be used alone to diagnose lung cancer, mediastinal involvement, or metastases. Confirmation with tissue biopsy is required. For brain metastases, magnetic resonance imaging (MRI) is the most effective method. MRI can also be useful in selected circumstances, such as superior sulcus tumors to rule out brachial plexus involvement, but in general, MRI does not play a major role in NSCLC staging.

In patients with NSCLC, the following are contraindications to potential curative resection: extrathoracic metastases, superior vena cava syndrome, vocal cord and, in most cases, phrenic nerve paralysis, malignant pleural effusion, cardiac tamponade, tumor within 2 cm of the carina (potentially curable with combined chemoradiotherapy), metastasis to the contralateral lung, metastases to supraclavicular lymph nodes, contralateral mediastinal node metastases (potentially curable with combined chemoradiotherapy), and involvement of the main pulmonary artery. In situations where it will make a difference in treatment, abnormal scan findings require tissue confirmation of malignancy so that patients are not precluded from having potentially curative therapy.

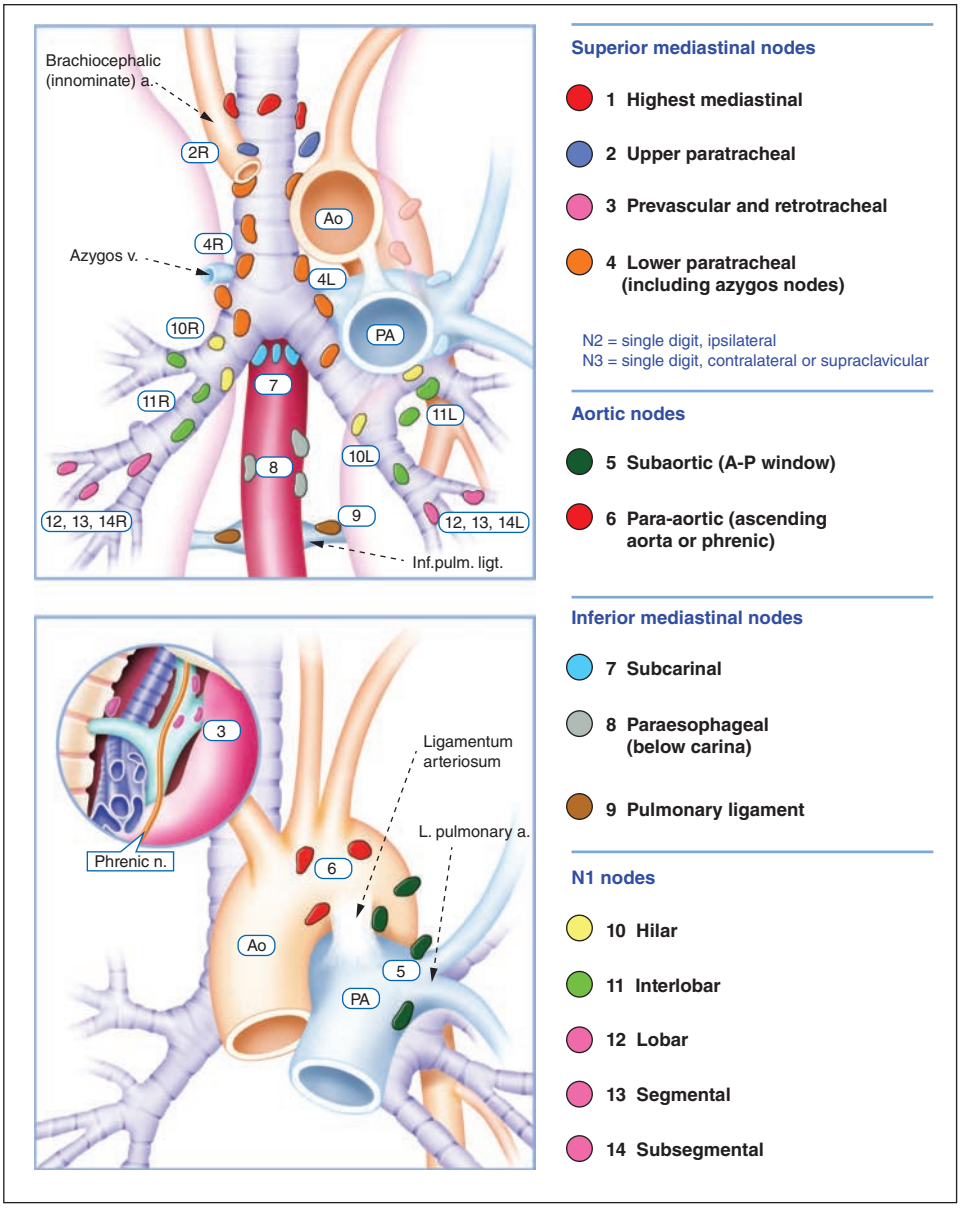


FIGURE 78-3 Lymph node stations in staging non-small-cell lung cancer. The International Association for the Study of Lung Cancer (IASLC) lymph node map, including the proposed grouping of lymph node stations into "zones" for the purposes of prognostic analyses. a., artery; Ao, aorta; Inf. pulm. ligt., inferior pulmonary ligament; n., nerve; PA, pulmonary artery; v., vein.

The best predictor of metastatic disease remains a careful history and physical examination. If signs, symptoms, or findings from the physical examination suggest the presence of malignancy, then sequential imaging starting with the most appropriate study should be performed. If the findings from the clinical evaluation are negative, then imaging studies beyond CT-PET are unnecessary and the search for metastatic disease is complete. In patients in whom distant metastatic disease has been ruled out, lymph node status needs to be assessed via minimally invasive techniques such as those mentioned above and/or invasive techniques such as mediastinoscopy, mediastinotomy, thoracoscopy, or thoracotomy. Approximately one-quarter to one-half of patients diagnosed with NSCLC will have mediastinal lymph node metastases at the time of diagnosis. Lymph node sampling is recommended in all patients with enlarged nodes detected by CT or PET scan and in patients with large tumors or tumors occupying the inner third of the lung. The extent of mediastinal lymph node involvement is important in determining the appropriate treatment strategy: surgical resection followed by adjuvant chemotherapy versus combined chemoradiation followed by immunotherapy (durvalumab) (see below). A standard nomenclature for referring to the location of lymph nodes involved with lung cancer has evolved **(Fig. 78-3)**.

In SCLC patients, current staging recommendations include a PET-CT scan and MRI of the brain (positive in 10% of asymptomatic patients). Bone marrow biopsies and aspirations are rarely performed now given the low incidence of isolated bone marrow metastases. Confirmation of metastatic disease, ipsilateral or contralateral lung nodules, or metastases beyond the mediastinum may be achieved by the same modalities recommended earlier for patients with NSCLC.

If a patient has signs or symptoms of spinal cord compression (pain, weakness, paralysis, urinary retention), a spinal CT or MRI scan should be performed. If metastases are evident on imaging, a neurosurgeon should be consulted for possible palliative surgical resection and/or a radiation oncologist should be consulted for palliative radiotherapy to the site of compression. If signs or symptoms of leptomeningeal disease develop at any time in a patient with lung cancer, an MRI of the brain and spinal cord should be performed, as well as a spinal tap, for

detection of malignant cells. If the spinal tap is negative, a repeat spinal tap should be considered. There is currently no approved therapy for the treatment of leptomeningeal disease.

STAGING SYSTEM FOR NON-SMALL-CELL LUNG CANCER

The tumor-node-metastasis (TNM) international staging system provides useful prognostic information and is used to stage all patients with NSCLC. The various T (tumor size), N (regional node involvement), and M (presence or absence of distant metastasis) stages are combined to form different stage groups (Tables 74-5 and 74-6). The eighth edition of the TNM staging system went into effect in 2018. T1 tumors are divided into tumors ≤1 cm (T1a), >1 cm and ≤2 cm (T1b), and >2 cm and ≤3 cm (T1c). T2 tumors are those that are >3 cm but ≤5 cm, involve the visceral pleura or main bronchus, or are associated with atelectasis; T2a tumors are >3 cm and ≤4 cm, and T2b are >4 cm and ≤5 cm. T3 tumors are >5 cm and ≤7 cm. T3 tumors also include tumors with invasion into local structures such as the chest wall and diaphragm and with additional nodules in the same lobe. T4 tumors include tumors >7 cm or tumors of any size with invasion into mediastinum, heart, great vessels, trachea, esophagus, or spine or with multiple nodules in the ipsilateral lung. Lymph node staging depends on metastasis to ipsilateral pulmonary or hilar nodes (N1), mediastinal or subcarinal nodes (N2), or contralateral mediastinal, hilar, or supraclavicular nodes (N3). Patients with metastasis may be classified as M1a (malignant pleural or pericardial effusion, pleural nodules, or nodules in the contralateral lung), M1b (single distant metastasis to a single organ; e.g., bone, liver, adrenal, or brain metastasis), or M1c (multiple metastases to a single organ or metastases to multiple organs). The effect of stage on survival is illustrated in Fig. 78-4. Approximately 15% of patients have localized disease that can be treated with curative attempt (surgery or radiotherapy), about a quarter have local or regional disease that may or may not be amenable to a curative attempt, and half have metastatic disease at the time of diagnosis. In 10%, the extent of disease is undefined.

TABLE 78-5 TNM Staging System for Lung Cancer (Eighth Edition)

Primary Tumor (T)	
T1	Tumor ≤3 cm diameter, surrounded by lung or visceral pleura, without invasion more proximal than lobar bronchus
T1mi	Minimally invasive adenocarcinoma (pure lepidic pattern, <3 cm in greatest dimension and <5 mm invasion)—T1a (size <1 cm)—T1b (1 cm < size <2 cm)—T1c (2 cm < size <3 cm)
T2	Tumor >3 cm but ≤7 cm, or tumor with any of the following features: Involves main bronchus ≥2 cm distal to carina Invades visceral pleura Associated with atelectasis or obstructive pneumonitis that extends to the hilar region but does not involve the entire lung
T2a	Tumor >3 cm but ≤5 cm
T2b	Tumor >5 cm but ≤7 cm
T3	Tumor >7 cm or any of the following: Directly invades any of the following: chest wall, diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium, main bronchus <2 cm from carina (without involvement of carina) Atelectasis or obstructive pneumonitis of the entire lung Separate tumor nodules in the same lobe
T4	Tumor of any size that invades the mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, or carina, or with separate tumor nodules in a different ipsilateral lobe
Nodal Stage (N)	
N0	No regional lymph node metastases
N1	Metastasis in ipsilateral peribronchial and/or ipsilateral hilar lymph nodes and intrapulmonary nodes, including involvement by direct extension
N2	Metastasis in ipsilateral mediastinal and/or subcarinal lymph node(s)
N3	Metastasis in contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene, or supraclavicular lymph node(s)
Metastases (M)	
M0	No distant metastasis
M1	Distant metastasis
M1a	Separate tumor nodule(s) in a contralateral lobe; tumor with pleural nodules or malignant pleural or pericardial effusion
M1b	Distant metastasis (in extrathoracic organs)
M1c	

Abbreviation: M1b, distant metastasis in single organ outside chest; M1c, multiple extrathoracic metastases to one or more organs; TNM, tumor-node-metastasis.

TABLE 78-6 TNM Stage Groupings, Eighth Edition

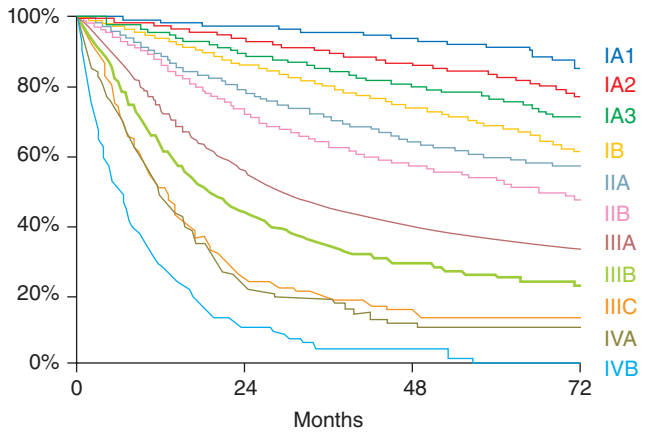
Stage IA1	T1a	N0	M0
Stage IA2	T1b	N0	M0
Stage IA3	T1c	N0	M0
Stage IB	T2a	N0	M0
Stage IIA	T2b	N0	M0
Stage IIB	T1a-T2b	N1	M0
	T3	N0	M0
Stage IIIA	T1-2b	N2	M0
	T3	N1	M0
	T4	N0/N1	M0
Stage IIIB	T1-2b	N3	M0
	T3/T4	N0/N1	M0
	T3/T4	N3	M0
Stage IVA	Any T	Any N	M1a/M1b
Stage IV B	Any T	Any N	M1c

STAGING SYSTEM FOR SMALL-CELL LUNG CANCER

In patients with SCLC, it is now recommended that both the Veterans Administration system and the American Joint Committee on Cancer/International Union Against Cancer eighth edition system (TNM) be used to classify the tumor stage. The Veterans Administration system is a distinct two-stage system dividing patients into those with limited- or extensive-stage disease. Patients with limited-stage disease (LD) have cancer that is confined to the ipsilateral hemithorax and can be encompassed within a tolerable radiation port. Thus, contralateral supraclavicular nodes, recurrent laryngeal nerve involvement, and superior vena caval obstruction can all be part of LD. Patients with extensive-stage disease (ED) have overt metastatic disease by imaging or physical examination. Cardiac tamponade, malignant pleural effusion, and bilateral pulmonary parenchymal involvement generally qualify disease as ED, because the involved organs cannot be encompassed safely or effectively within a single radiation therapy port. Sixty to 70% of patients are diagnosed with ED at presentation. The TNM staging system is preferred in the rare SCLC patient presenting with what appears to be clinical stage I disease (see above).

PHYSIOLOGIC STAGING

Patients with lung cancer often have other comorbid conditions related to smoking including cardiovascular disease and COPD. To improve their preoperative condition, correctable problems (e.g., anemia, electrolyte and fluid disorders, infections, cardiac disease, and arrhythmias) should be addressed, appropriate chest physical therapy should be instituted, and patients should be encouraged to stop smoking. Patients with a forced expiratory volume in 1 s (FEV_1) of >2 L or >80% of predicted can tolerate a pneumonectomy, and those with an FEV_1 >1.5 L have adequate reserve for a lobectomy. In patients with borderline lung function but a resectable tumor, cardiopulmonary exercise testing could be performed as part of the physiologic evaluation. This test allows an estimate of the maximal oxygen consumption (Vo_{2max}). A Vo_{2max}



Stage	24 months	60 months
IA1	97%	92%
IA2	94%	83%
IA3	90%	77%
IB	87%	68%
IIA	79%	60%
IIB	72%	53%
IIIA	55%	36%
IIIB	44%	26%
IIIC	24%	13%
IVA	23%	10%
IVB	10%	0%

FIGURE 78-4 Influence of non-small-cell lung cancer stage on survival.

<15 mL/(kg.min) predicts for a higher risk of postoperative complications. Patients deemed unable to tolerate lobectomy or pneumonectomy from a pulmonary functional standpoint may be candidates for more limited resections, such as wedge or anatomic segmental resection, although such procedures are associated with significantly higher rates of local recurrence and a trend toward decreased overall survival. All patients should be assessed for cardiovascular risk using American College of Cardiology and American Heart Association guidelines. A myocardial infarction within the past 3 months is a contraindication to thoracic surgery because 20% of patients will die of reinfarction. An infarction in the past 6 months is a relative contraindication. Other major contraindications include uncontrolled arrhythmias, an FEV_1 of <1 L, CO_2 retention (resting P_{CO_2} >45 mmHg), DLCO <40%, and severe pulmonary hypertension.

TREATMENT

Non-Small-Cell Lung Cancer

The overall treatment approach to patients with NSCLC is shown in **Fig. 78-5**.

OCCULT AND STAGE 0 CARCINOMAS

Patients with severe atypia on sputum cytology have an increased risk of developing lung cancer compared to those without atypia. In the uncommon circumstance where malignant cells are identified in a sputum or bronchial washing specimen but the chest imaging appears normal (TX tumor stage), the lesion must be localized. More than 90% of tumors can be localized by meticulous examination of the bronchial tree with a fiberoptic bronchoscope under general anesthesia and collection of a series of differential brushings and biopsies. Surgical resection following bronchoscopic localization has been shown to improve survival compared to no treatment. Close follow-up of these patients is indicated because of the high incidence of second primary lung cancers (5% per patient per year).

SOLITARY PULMONARY NODULE AND "GROUND-GLASS" OPACITIES

A solitary pulmonary nodule is defined as an x-ray density completely surrounded by normal aerated lung with circumscribed margins, of any shape, usually 1–6 cm in greatest diameter. The approach to a patient with a solitary pulmonary nodule is based on an estimate of the probability of cancer, determined according to the patient's smoking history, age, and characteristics on imaging **(Table 78-7)**. Prior CXRs and CT scans should be obtained if available for comparison. A PET scan may be useful if the lesion is greater than 7–8 mm in diameter. If no diagnosis is apparent, Mayo investigators reported that clinical characteristics (age, cigarette smoking status, and prior cancer diagnosis) and three radiologic characteristics (nodule diameter, spiculation, and upper lobe location) were independent predictors of malignancy. At present, only two radiographic criteria are thought to predict the benign nature of a solitary pulmonary nodule: lack of growth over a period >2 years and certain characteristic patterns of calcification. Calcification alone, however, does not exclude malignancy; a dense central nidus, multiple punctuate foci, and "bull's eye" (granuloma) and "popcorn ball" (hamartoma) calcifications are highly suggestive of a benign lesion. In contrast, a relatively large lesion, lack of or asymmetric calcification, chest symptoms, associated atelectasis, pneumonitis, or growth of the lesion revealed by comparison with an old x-ray or CT scan or a positive PET scan may be suggestive of a malignant process and warrant further attempts to establish a histologic diagnosis. An algorithm for assessing these lesions is shown in **Fig. 78-6**.

Since the advent of screening CTs, small "ground-glass" opacities (GGOs) have often been observed, particularly as the increased sensitivity of CTs enables detection of smaller lesions. Many of these GGOs, when biopsied, are found to be atypical adenomatous hyperplasia (AAH), adenocarcinoma in situ (AIS), or minimally invasive adenocarcinoma (MIA). AAH is usually a nodule of <5 mm and is minimally hazy, also called nonsolid or ground glass (i.e., hazy slightly increased attenuation, no solid component, and preservation of bronchial and vascular margins). On thin-section CT, AIS is usually a nonsolid nodule and tends to be slightly more opaque than AAH. MIA is mainly solid, usually with a small (<5 mm) central solid component. However, overlap exists among the imaging features of the preinvasive and minimally invasive lesions in the lung adenocarcinoma spectrum. Lepidic adenocarcinomas are usually solid but may be nonsolid. Likewise, the small invasive adenocarcinomas also are usually solid but may exhibit a small nonsolid component.

MANAGEMENT OF STAGES I AND II NSCLC

Surgical Resection of Stage I and II NSCLC Surgical resection, ideally by an experienced thoracic surgeon, is the treatment of choice for patients with clinical stage I and II NSCLC who are able to tolerate the procedure. Operative mortality rates for patients resected by thoracic or cardiothoracic surgeons are lower compared to general surgeons. Moreover, survival rates are higher in patients who undergo resection in facilities with a high surgical volume compared to those performing fewer than 70 procedures per year, even though the higher-volume facilities often serve older and less socioeconomically advantaged populations. The improvement in survival is most evident in the immediate postoperative period. In patients with stage I NSCLC, lobectomy is superior to wedge

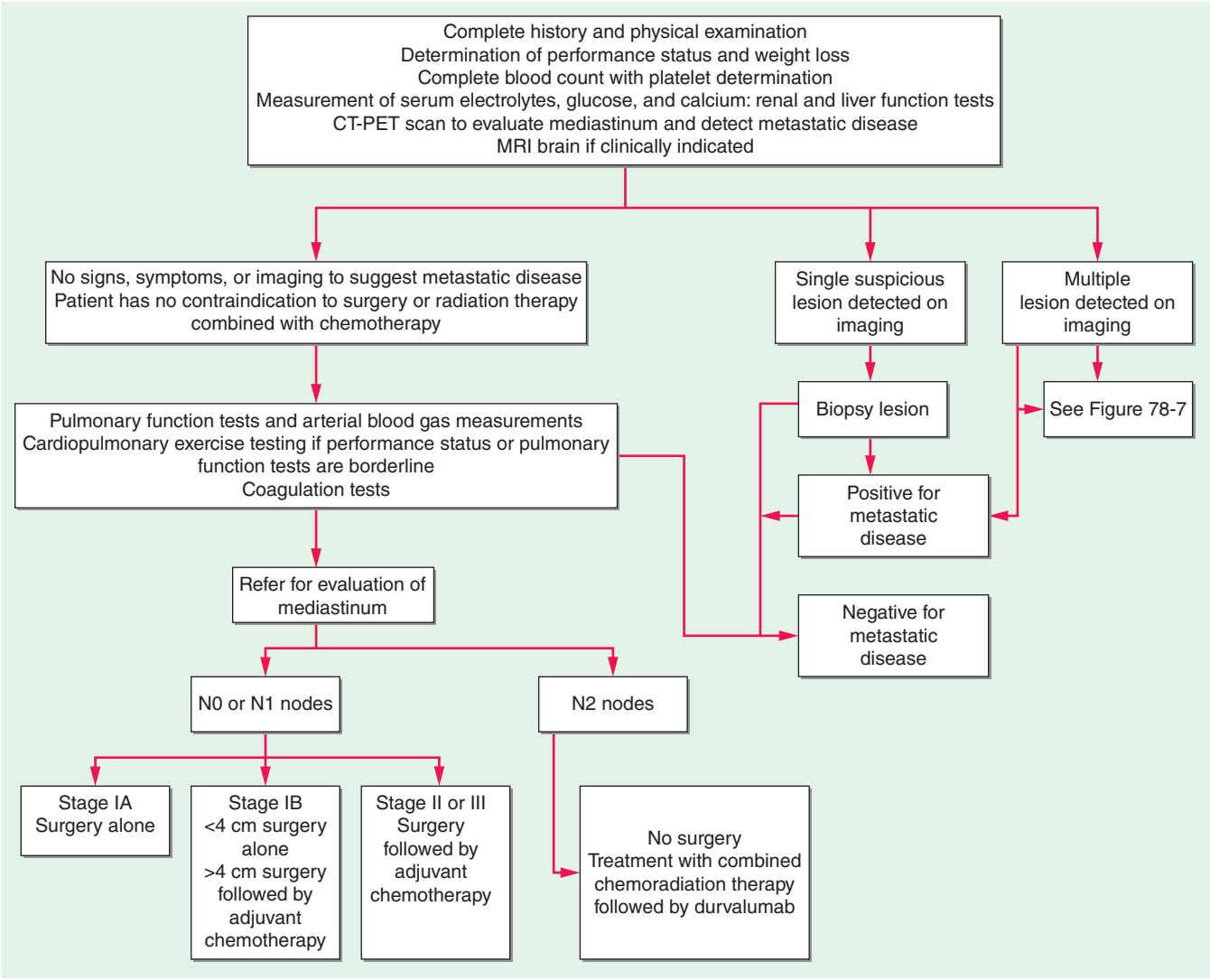


FIGURE 78-5 Algorithm for management of non-small-cell lung cancer. MRI, magnetic resonance imaging; PET, positron emission tomography.

resection with respect to rates of local recurrence. There is also a trend toward improvement in overall survival. In patients with comorbidities, compromised pulmonary reserve, and small peripheral lesions, a limited resection, wedge resection, or segmentectomy (potentially by video-assisted thoracoscopic surgery) may be reasonable surgical options. Pneumonectomy is reserved for patients with central tumors and should be performed only in patients with excellent pulmonary reserve. The 5-year survival rates are 68–92% for patients with stage I NSCLC and 53–60% for patients with stage II NSCLC.

TABLE 78-7 Assessment of Risk of Cancer in Patients with Solitary Pulmonary Nodules

	RISK		
VARIABLE	LOW	INTERMEDIATE	HIGH
Diameter (cm)	<1.5	1.5–2.2	≥2.3
Age (years)	<45	45–60	>60
Smoking status	Never smoker	Current smoker (<20 cigarettes/d)	Current smoker (>20 cigarettes/d)
Smoking cessation status	Quit ≥7 years ago or quit	Quit <7 years ago	Never quit
Characteristics of nodule margins	Smooth	Scalloped	Corona radiata or spiculated

Source: From D Ost et al: The solitary pulmonary nodule. N Engl J Med 348:2535, 2003. Copyright © (2003) Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

Accurate pathologic staging requires adequate segmental, hilar, and mediastinal lymph node sampling. Ideally, this includes a mediastinal lymph node dissection. On the right side, mediastinal stations 2R, 4R, 7, 8R, and 9R should be dissected; on the left side, stations 5, 6, 7, 8L, and 9L should be dissected. Hilar lymph nodes are typically resected and sent for pathologic review, although it is helpful to specifically dissect and label level 10 lymph nodes when possible. On the left side, level 2 and sometimes level 4 lymph nodes are generally obscured by the aorta. Although the therapeutic benefit of nodal dissection versus nodal sampling is controversial, a pooled analysis of three trials involving patients with stages I to IIIA NSCLC demonstrated a superior 4-year survival in patients undergoing resection and a complete mediastinal lymph node dissection compared with lymph node sampling. Moreover, complete mediastinal lymphadenectomy added little morbidity to a pulmonary resection for lung cancer when carried out by an experienced thoracic surgeon.

Radiation Therapy in Stages I and II NSCLC There is currently no role for postoperative radiation therapy in patients following resection of stage I or II NSCLC with negative margins. However, patients with stage I and II disease who either refuse or are not suitable candidates for surgery should be considered for radiation therapy with *curative* intent. Stereotactic body radiation therapy (SBRT) is a technique used to treat patients with isolated pulmonary nodules (≤5 cm) who are not candidates for or refuse surgical resection. Treatment is typically administered in three to five

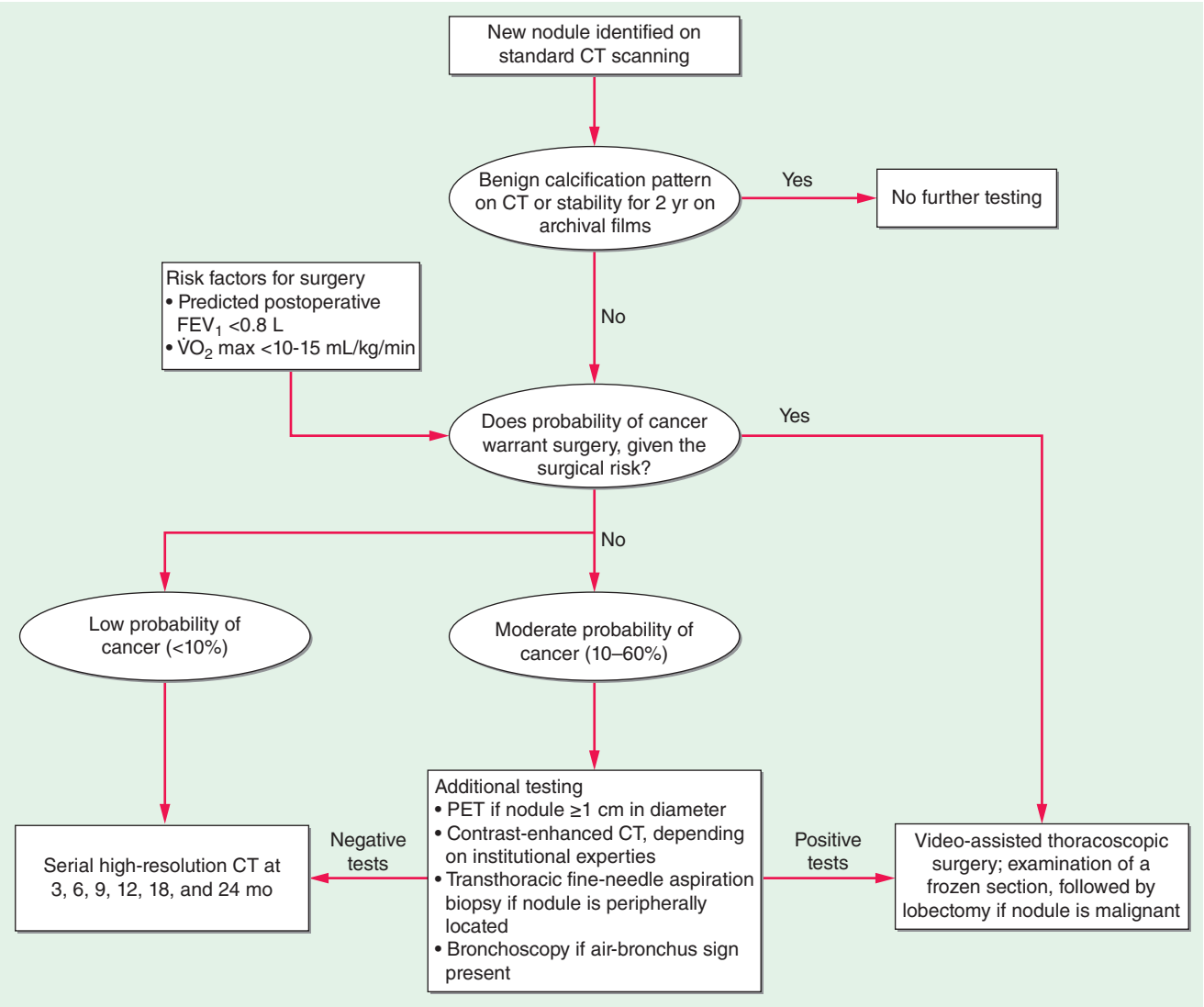


FIGURE 78-6 Approach to the solitary pulmonary nodule. FEV_1, forced expiratory volume in 1 s; PET, positron emission tomography.

fractions delivered over 1–2 weeks. In uncontrolled studies, disease control rates are >90%, and 5-year survival rates of up to 60% have been reported with SBRT. By comparison, survival rates typically range from 13 to 39% in patients with stage I or II NSCLC treated with standard external-beam radiotherapy. Cryoablation is another technique occasionally used to treat small, isolated tumors (i.e., ≤3 cm). However, very little data exist on long-term outcomes with this technique.

Chemotherapy in Stages I and II NSCLC Although a landmark meta-analysis of cisplatin-based adjuvant chemotherapy trials in patients with resected stages I to IIIA NSCLC (the Lung Adjuvant Cisplatin Evaluation [LACE] Study) demonstrated a 5.4% improvement in 5-year survival for adjuvant chemotherapy compared to surgery alone, the survival benefit was seemingly confined to patients with stage II or III disease **(Table 78-8)**. By contrast, survival was actually worsened in stage IA patients with the application of adjuvant therapy. In stage IB, there was a modest improvement in survival of questionable clinical significance. Adjuvant chemotherapy was also detrimental in patients with poor performance status (Eastern Cooperative Oncology Group [ECOG] performance status = 2). These data suggest that adjuvant chemotherapy is best applied in patients with resected stage II or III NSCLC. There is no apparent role for adjuvant chemotherapy in patients with resected stage IA or IB NSCLC. A possible exception to the prohibition of adjuvant therapy in this setting is the stage IB patient with a resected lesion ≥4 cm. Osimertinib, an EGFR TKI, demonstrated improved disease-free survival for patients with *EGFR* mutation (exon 19 deletion or L858R)–positive NSCLC treated for 3 years following chemotherapy. However, the effect on overall survival is unknown.

As with any treatment recommendation, the risks and benefits of adjuvant chemotherapy should be considered on an individual patient basis. If a decision is made to proceed with adjuvant chemotherapy, in general, treatment should be initiated 6–12 weeks after surgery, assuming the patient has fully recovered, and should be administered for no more than four cycles. Although cisplatin-based chemotherapy is the preferred treatment regimen, carboplatin can be substituted for cisplatin in patients who are unlikely to tolerate cisplatin for reasons such as reduced renal function, presence of neuropathy, or hearing impairment. A large cooperative group trial compared cisplatin-based chemotherapy with vinorelbine, pemetrexed, gemcitabine, or docetaxel with or without antiangiogenic therapy. While adding antiangiogenic therapy to platinum-based chemotherapy offered no benefit, the trial also demonstrated no difference in survival among the four chemotherapy regimens. Therefore, no specific chemotherapy regimen is considered superior in this setting, and treatment selection may be based on cost and patient comorbidities.

Neoadjuvant chemotherapy, which is the application of chemotherapy administered *before* an attempted surgical resection, has been advocated by some experts on the assumption that such an approach will more effectively extinguish occult micrometastases compared to

TABLE 78-8 Adjuvant Chemotherapy Trials in Non-Small-Cell Lung Cancer

TRIAL	STAGE	TREATMENT	NO. OF PATIENTS	5-YEAR SURVIVAL (%)	*P*
IALT	I–III	Cisplatin-based	932	44.5	< .03
		Control	835	40.4	
BR10	IB–II	Cisplatin + vinorelbine	242	69	.03
		Control	240	54	
ANITA	IB–IIIA	Cisplatin + vinorelbine	407	60	.017
		Control	433	58	
ALPI	I–III	MVP	548	50	.49
		Control	540	45	
BLT	I–III	Cisplatin-based	192	60	.90
		Control	189	58	
CALGB	IB	Carboplatin + paclitaxel	173	59	.10
			171	57	
ECOG1505	IB > 4c – IIIA	Cisplatin – based	749	NR	.90
		Cisplatin – based + bevacizumab	752	NR	

Abbreviations: ALPI, Adjuvant Lung Cancer Project Italy; ANITA, Adjuvant Navelbine International Trialist Association; BLT, Big Lung Trial; CALGB, Cancer and Lung Cancer Group B; ECOG, Eastern Cooperative Oncology Group; IALT, International Adjuvant Lung Cancer Trial; MVP, mitomycin, vindesine, and cisplatin; NR, not reported.

postoperative chemotherapy. In addition, it is thought that preoperative chemotherapy might render an inoperable lesion resectable. A meta-analysis of 15 randomized controlled trials involving more than 2300 patients with stage I to III NSCLC suggested there may be a modest 5-year survival benefit (i.e., ~5%) that is virtually identical to the survival benefit achieved with postoperative chemotherapy. Accordingly, neoadjuvant therapy may prove useful in selected cases (see below). A decision to use neoadjuvant chemotherapy should always be made in consultation with an experienced surgeon.

All patients with resected NSCLC are at high risk of developing a second primary lung cancer or recurrence, most of which occur within 18–24 months of surgery. Thus, it is reasonable to follow these patients with periodic imaging studies. Given the results of the NLST, periodic CT scans appear to be the most appropriate screening modality. Based on the timing of most recurrences, some guidelines recommend a contrasted chest CT scan every 6 months for the first 3 years after surgery, followed by yearly CT scans of the chest without contrast thereafter.

MANAGEMENT OF STAGE III NSCLC

Management of patients with stage III NSCLC usually requires a combined-modality approach. Patients with stage IIIA disease commonly are stratified into those with "nonbulky" or "bulky" mediastinal lymph node (N2) disease. Although the definition of "bulky" N2 disease varies somewhat in the literature, the usual criteria include the size of a dominant lymph node (i.e., >2–3 cm in short-axis diameter as measured by CT), groupings of multiple smaller lymph nodes, evidence of extracapsular nodal involvement, or involvement of more than two lymph node stations. The distinction between nonbulky and bulky stage IIIA disease is mainly used to select potential candidates for *upfront* surgical resection or for resection after neoadjuvant therapy. Many aspects of therapy of patients with stage III NSCLC remain controversial, and the optimal treatment strategy has not been clearly defined. Furthermore, because stage III disease is highly heterogeneous, no single treatment approach can be recommended for all patients. Key factors guiding treatment choices include the particular combination of tumor (T) and nodal (N) disease, the ability to achieve a complete surgical resection if indicated, and the patient's overall physical condition and preferences. For example, in carefully selected patients with limited stage IIIA disease where involved mediastinal lymph nodes can be completed resected, initial surgery followed by postoperative chemotherapy (with or without radiation therapy) may be indicated. By contrast, for patients with clinically evident bulky mediastinal lymph node involvement, the standard approach to treatment is concurrent chemoradiotherapy followed by a year of immunotherapy with durvalumab or other PD-L1-directed antibody.

Absent and Nonbulky Mediastinal (N2, N3) Lymph Node Disease

For the subset of stage IIIA patients initially thought to have clinical stage I or II disease (i.e., pathologic involvement of mediastinal [N2] lymph nodes is *not* detected preoperatively), surgical resection is often the treatment of choice. This is followed by adjuvant chemotherapy in patients with microscopic lymph node involvement in a resection specimen. Postoperative radiation therapy (PORT) may also have a role for those with close or positive surgical margins. Patients with tumors exceeding 7 cm in size or involving the chest wall or proximal airways within 2 cm of the carina with hilar lymph node involvement (but not N2 disease) are classified as having T3N1 stage IIIA disease. They too are best managed with surgical resection, if technically feasible, followed by adjuvant chemotherapy if completely resected. Patients with T3N0 or T3N1 disease due to the presence of satellite nodules within the same lobe as the primary tumor are also candidates for surgery, as are patients with ipsilateral nodules in another lobe and negative mediastinal nodes (IIIA, T4N0 or T4N1). Although data regarding adjuvant chemotherapy in the latter subsets of patients are limited, it is often recommended.

Patients with T4N0-1 may have involvement of the carina, superior vena cava, or a vertebral body and yet still be candidates for surgical resection in selected circumstances. The decision to proceed with an attempted resection must be made in consultation with an experienced thoracic surgeon often in association with a vascular or cardiac surgeon and an orthopedic surgeon depending on tumor location. However, if an incomplete resection is inevitable or if there is evidence of N2 involvement (stage IIIB), surgery for T4 disease is contraindicated. Most T4 lesions are best treated with concurrent chemoradiotherapy followed by durvalumab.

The role of PORT in patients with completely resected stage III NSCLC is controversial. To a large extent, the use of PORT is dictated by the presence or absence of N2 involvement and, to a lesser degree, by the biases of the treating physician. Using the Surveillance, Epidemiology, and End Results (SEER) database, a recent meta-analysis of PORT identified a significant increase in survival in patients with N2 disease but not in patients with N0 or N1 disease. An earlier analysis by the PORT Meta-analysis Trialist Group using an older database produced similar results.

Known Mediastinal (N2, N3) Lymph Node Disease

When pathologic involvement of mediastinal lymph nodes is documented preoperatively, a combined-modality approach is recommended assuming the patient is a candidate for treatment with curative intent. These patients are at high risk for both local and distant recurrence if managed with resection alone. For patients with stage III disease who are not candidates for surgical resection, *concurrent* chemoradiotherapy is most commonly used as the initial treatment followed by durvalumab. Concurrent chemoradiotherapy has been shown to produce superior survival compared to *sequential* chemoradiotherapy; however, it also is associated with greater host toxicities (including fatigue, esophagitis, and neutropenia). Therefore, for patients with a good performance status, concurrent chemoradiotherapy is the preferred treatment approach, whereas sequential chemoradiotherapy may be more appropriate for patients with a performance status that is not as good. For patients who are *not* candidates for a combined-modality treatment approach, typically due to a poor performance status or a comorbidity that makes chemotherapy untenable, radiotherapy alone may provide a modest survival benefit in addition to symptom palliation.

For patients with potentially resectable N2 disease, it remains uncertain whether surgery after neoadjuvant chemoradiotherapy improves survival. In an NCI-sponsored Intergroup randomized trial comparing concurrent chemoradiotherapy alone to concurrent chemoradiotherapy followed by attempted surgical resection, no survival benefit was observed in the trimodality arm compared to the bimodality therapy. In fact, patients subjected to a pneumonectomy had a worse survival outcome. By contrast, those treated with a lobectomy appeared to have a survival advantage based on a retrospective subset analysis. Thus, in carefully selected, otherwise healthy patients with nonbulky mediastinal lymph node involvement, surgery may be a reasonable option if the primary tumor can be fully resected with a lobectomy. This is not the case if a pneumonectomy is required to achieve complete resection.

Superior Sulcus Tumors (Pancoast Tumors) Superior sulcus tumors represent a distinctive subset of stage III disease. These tumors arise in the apex of the lung and may invade the second and third ribs, the brachial plexus, the subclavian vessels, the stellate ganglion, and adjacent vertebral bodies. They also may be associated with Pancoast syndrome, characterized by pain that may arise in the shoulder or chest wall or radiate to the neck. Pain characteristically radiates to the ulnar surface of the hand. Horner's syndrome (enophthalmos, ptosis, miosis, and anhidrosis) due to invasion of the paravertebral sympathetic chain may be present as well. Patients with these tumors should undergo the same staging procedures as all patients with stage II and III NSCLC. Neoadjuvant chemotherapy or combined chemoradiotherapy followed by surgery is reserved for those without N2 involvement. This approach yields excellent survival outcomes (>50% 5-year survival in patients with an R0 resection). Patients with N2 disease are less likely to benefit from surgery and can be managed with chemoradiotherapy followed by durvalumab. Patients presenting with metastatic disease can be treated with radiation therapy (with or without chemotherapy) for symptom palliation.

MANAGEMENT OF METASTATIC NSCLC

Approximately 40% of NSCLC patients present with advanced, stage IV disease at the time of diagnosis. In addition, a significant number of patients who first presented with early-stage NSCLC will eventually relapse with distant disease. Patients who have recurrent disease have a better prognosis than those presenting with metastatic disease at the time of diagnosis. Standard medical management, the judicious use of pain medications, and the appropriate use of radiotherapy and systemic therapy—which may consist of targeted therapy, immunotherapy, and/or traditional cytotoxic chemotherapy depending on the specific diagnosis as well as PD-L1 tumor proportion score (TPS) and molecular subtype—form the cornerstone of management. Systemic therapy palliates symptoms, improves quality of life, and improves survival in patients with metastatic NSCLC, particularly in patients with good performance status. Of note, the early application of palliative care in conjunction with chemotherapy in patients with advanced NSCLC is associated with both improved survival and quality of life.

Targeted Therapies for Select Molecular Cohorts of NSCLC For a cohort of NSCLC patients, the presence of an oncogenic driver mutation allows the use of oral therapies with significant antitumor activity and improved survival compared to cytotoxic chemotherapy. These driver mutations occur in genes encoding signaling proteins that, when aberrant, promote the uncontrolled growth and metastasis of tumor cells. Importantly, driver mutations can serve as Achilles' heels for tumors, if their gene products can be targeted therapeutically with small-molecule inhibitors. All patients with advanced NSCLC should undergo molecular testing with broad panel-based testing techniques such as next-generation sequencing (NGS) to look for oncogenic drivers. Mutations, fusions, and deletions have been reported in a number of genes including *EGFR*, *ALK*, *ROS1*, *BRAF*, *RET*, *MET*, *NTRK*, *KRAS*, *PIK3CA*, *NRAS*, *AKT1*, and *MEK1* (*MAP2K1*); however, not all are considered actionable at this time. As our treatment armamentarium expands, knowledge of these mutations is critical for selection of appropriate therapy.

EGFR mutations have been detected in 10–15% of North American patients diagnosed with NSCLC. *EGFR* mutations are associated with younger age, light (<10 pack-year) and nonsmokers, and adenocarcinoma histology. Approximately 90% of these mutations are exon 19 deletions or exon 21 L858R point mutations within the EGFR TK domain, resulting in hyperactivation of both EGFR kinase activity and downstream signaling. Lung tumors that harbor activating mutations within the EGFR kinase domain display high sensitivity to small-molecule EGFR TKIs. Osimertinib, erlotinib, gefitinib, afatinib and dacomitinib are FDA-approved oral small-molecule TKIs that inhibit EGFR. Several large, international, phase 3 studies have demonstrated improved response rates and progression-free survival in patients with *EGFR* mutation–positive NSCLC treated with an EGFR TKI as compared with standard first-line chemotherapy regimens **(Table 78-9)**. Osimertinib was shown in a randomized phase 3 trial to have superior progression-free and overall survival in patients with *EGFR*-mutant NSCLC compared to earlier-generation EGFR TKIs (erlotinib or gefitinib) and to chemotherapy.

Chromosomal rearrangements involving the anaplastic lymphoma kinase (*ALK*) gene on chromosome 2 have been found in ~3–7% of patients with NSCLC. *ALK* rearrangements lead to hyperactivation of the ALK TK domain. Similar to EGFR, *ALK* rearrangements are typically (but not exclusively) associated with younger age, light (<10 pack-year) and nonsmokers, and adenocarcinoma histology. Crizotinib is a first-generation ALK TKI, whereas alectinib, brigatinib and ceritinib are second-generation ALK TKIs approved as first-line treatment options for patients with lung tumors harboring *ALK* rearrangements. Both alectinib and brigatinib have been found to have superior progression-free survival to crizotinib, whereas lorlatinib, a third-generation ALK TKI, is approved in patients who progress on a second-generation ALK TKI **(Table 78-10)**. *ALK* testing may be performed via fluorescence

TABLE 78-9 Phase 3 Trials of EGFR TKIs in EGFR-Positive Non-Small-Cell Lung Cancer

TRIAL	THERAPY	NO. OF PATIENTS	ORR (%)	PFS (MONTHS)
IPASS	CbP	129	47	6.3
	Gefitinib	132	71	9.3
EURTAC	CG	87	15	5.2
	Erlotinib	86	58	9.7
OPTIMAL	CG	72	36	4.6
	Erlotinib	82	83	13.1
NEJ002	CG	114	31	5.4
	Gefitinib	114	74	10.8
WJTOG3405	CD	89	31	6.3
	Gefitinib	88	62	9.2
LUX LUNG 3	CP	115	23	6.9
	Afatinib	230	56	11.1
LUX LUNG 6	CG	122	23	5.6
	Afatinib	242	67	11.0
LUX LUNG 7	Erlotinib	159	58	10.9
	Afatinib	160	73	11.0
ARCHER 1050	Gefitinib	225	72	9.2
	Dacomitinib	227	75	14.7
FLAURA	Erlotinib or Gefitinib	277	76	8.5
	Osimertinib	279	80	17.2

Abbreviations: CbP, carboplatin and paclitaxel; CD, cisplatin and docetaxel; CG, cisplatin and gemcitabine; CP, cisplatin and paclitaxel; ORR, overall response rate; PFS, progression-free survival.

TABLE 78-10 Results of Phase 3 Trials Comparing First-Line ALK Inhibitors in ALK-Positive NSCLC

TRIAL	THERAPY	NO. OF PATIENTS	ORR (%)	PFS (MONTHS)
Profile 1014	Crizotinib	172	74	10.9
	Platinum-chemotherapy	171	45	7.0
ALEX	Alectinib	152	82.9	25.7
	Crizotinib	151	75.5	10.4
J-ALEX	Alectinib	103	92	34.1
	Crizotinib	104	79	10.2
ALTA1L	Brigatinib	137	71	67% at 1 year
	Crizotinib	138	60	43% at 1 year

Abbreviations: NSCLC, non-small-cell lung cancer; ORR, overall response rate; PFS, progression-free survival.

in situ hybridization (FISH), immunohistochemistry (IHC), or NGS.

ROS1 fusions, detected by FISH or NGS, have been identified in ~1% of patients with NSCLC, and similar to *EGFR* mutations and *ALK* fusions, *ROS1* rearrangements are typically associated with younger age and light or never smoking status. Crizotinib and lorlatinib, which inhibit both ALK and ROS1 kinases, and entrectinib have been FDA approved for patients whose tumors harbor a *ROS1* fusion.

NTRK fusions occur in members of the *NTRK* gene family (*NTRK1*, *NTRK2*, *NTRK3*) and result in constitutive protein kinase activation. *NTRK* fusions are rare, occurring in <1% of patients with NSCLC. Similar to the above mutations, they more commonly occur in never smokers; however, patients with *NTRK* fusions are often older patient compared to those with *ROS1* and *ALK* alterations. Entrectinib and larotrectinib have demonstrated durable antitumor efficacy and are currently approved for *NTRK*-positive NSCLC.

MET exon 14 skipping mutations have also been identified in approximately 3–5% of patients with NSCLC. Unlike the above-mentioned mutations, *MET* exon 14 skipping mutations occur in both squamous and nonsquamous NSCLC patients and those with a history of smoking. Pharmacologic inhibition of the overactive MET pathway with capmatinib or tepotinib resulted response rates >70%, particularly in treatment-naïve NSCLC patients.

Oncogenic mutations in *BRAF* have been observed in ~2% of patients with NSCLC and, similar to *MET*, occur in both squamous and nonsquamous NSCLC and with an equal prevalence in patients with a history of smoking. This mutation is typically most targetable when it occurs at the 600th amino acid valine (V600). Combined inhibition with a BRAF and MEK inhibitor, dabrafenib plus trametinib, is a first-line or later therapeutic option in patients with *BRAF* V600–mutant NSCLC and appears to be superior to BRAF or MEK inhibition alone.

RET alterations typically occur as chromosomal rearrangements resulting in constitutive TKI activation. *RET* rearrangements may be detected by either FISH or NGS in ~1% of NSCLC patients. Analogous to capmatinib, selpercatinib has demonstrated an excellent response rate; as many as 85% of treatment-naïve NSCLC patients with *RET* alterations responded. All National Comprehensive Cancer Network–supported targetable oncogenic driver mutations and potential therapeutic options are summarized in Fig. 78-7.

Mutations within the *KRAS* GTPase are found in ~20% of lung adenocarcinomas. Agents targeting *KRAS* G12C are in development. Each of the other driver mutations occurs in <1–3% of lung adenocarcinomas. The great majority of the driver mutations are mutually exclusive. Most cancers have just one main driver. Defining mechanisms of acquired resistance to small-molecule inhibitors is a high research priority.

Immunotherapy Immune checkpoint inhibitors have significantly improved the quality of life and survival for a group of patients with locally advanced and metastatic NSCLC. These agents are used primarily in patients whose tumors do not express a targetable genetic lesion (Fig. 78-8). Immune checkpoint inhibitors work by blocking interactions between T cells and antigen presenting cells (APCs) or tumor cells that lead to T-cell inactivation. By inhibiting this interaction, the immune system is effectively upregulated and T cells become activated against tumor cells. Several randomized studies have demonstrated superior overall survival in patients treated with pembrolizumab or atezolizumab monotherapy or nivolumab plus ipilimumab combination immunotherapy compared to chemotherapy in patients with metastatic NSCLC with PD-L1 expression in ≥50% of tumor cells (Keynote 024, IMPOWER 110) and ≥1% of tumor cells (Keynote 042, CheckMate 227) (Table 78-11). The evidence supporting the use of single-agent immunotherapy in patients with tumor PD-L1 <50% remains unclear; current recommendations suggest the use of chemotherapy plus immunotherapy or immunotherapy combinations as the first-line treatment strategy in patients with metastatic NSCLC with tumor PD-L1 <50%. As discussed below, specific regimens vary by tumor histology (adenocarcinoma vs squamous cell carcinoma). Although PD-L1 has been identified as a biomarker that can predict response to immune checkpoint inhibitors, responses are observed in patients who do

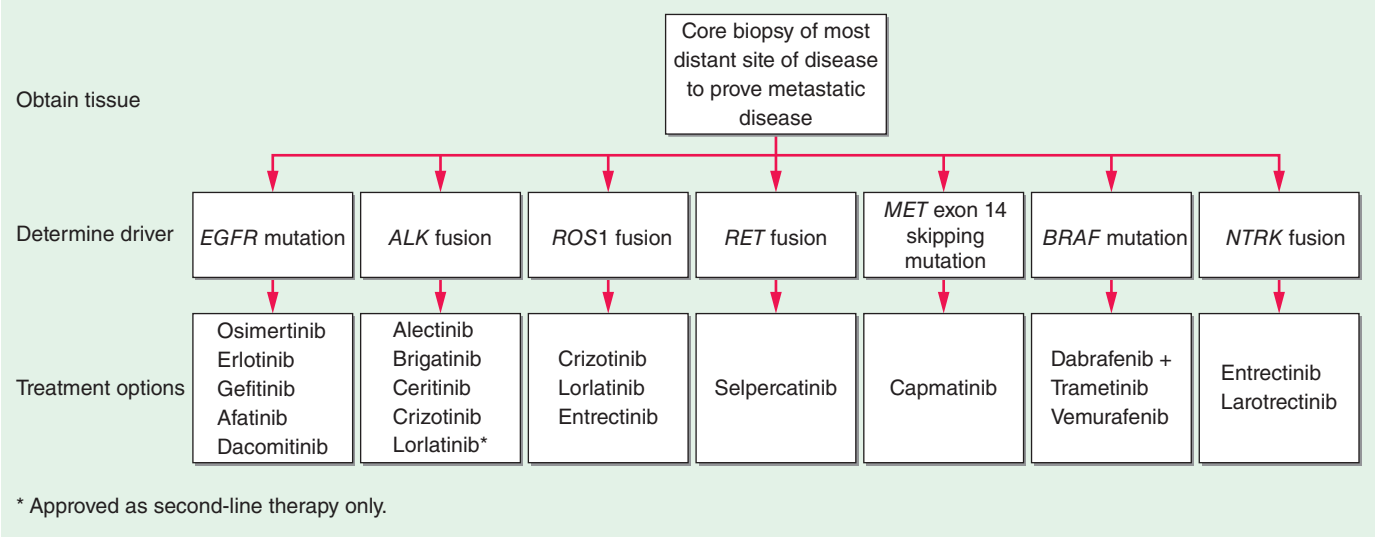


FIGURE 78-7 Approach to targeted therapy in non-small-cell lung cancer (NSCLC).

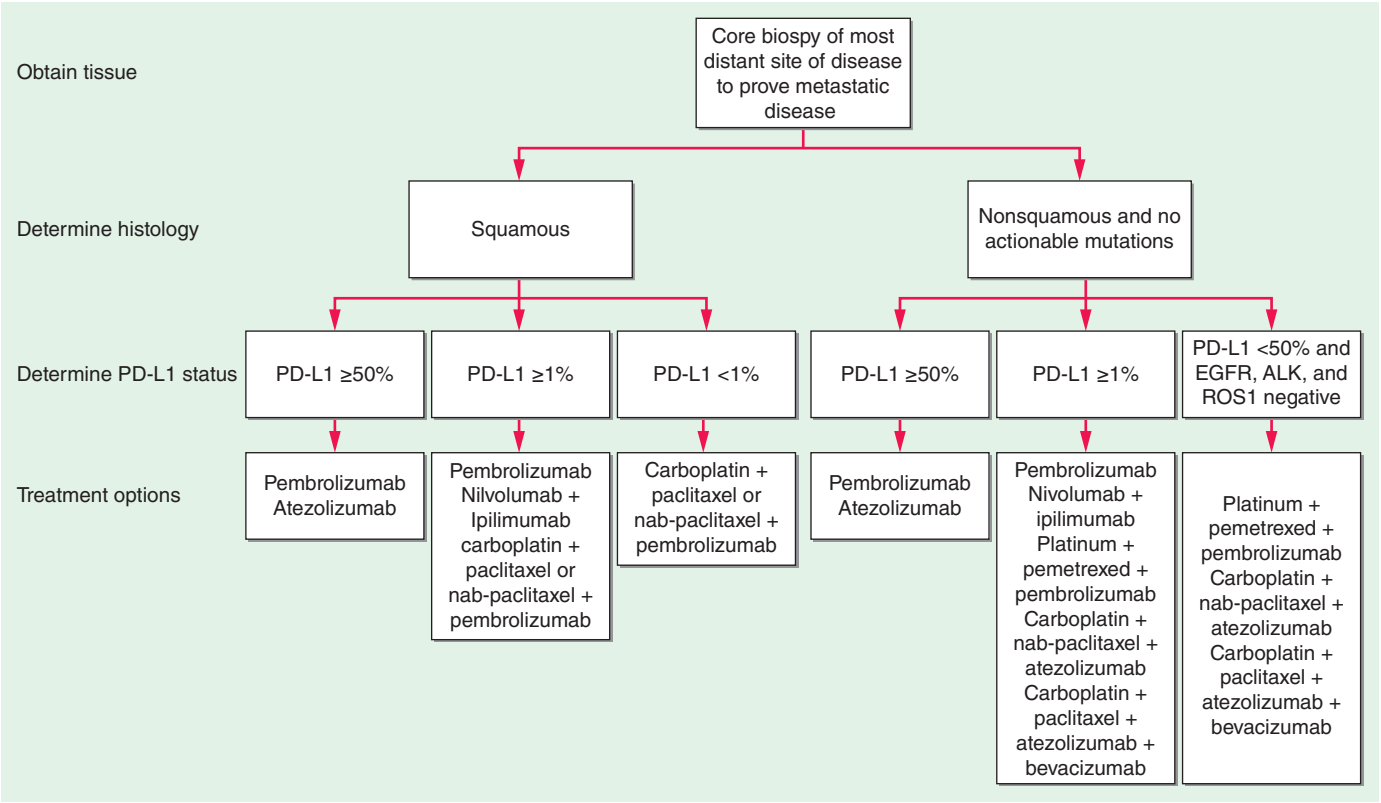


FIGURE 78-8 Approach to first-line therapy in a patient with stage IV, driver mutation–negative non-small-cell lung cancer (NSCLC).

not appear to express the biomarker, and not all PD-L1-positive patients respond to checkpoint inhibition. Importantly patients with driver mutations such as *EGFR* and *ALK* appear to derive greater benefit from targeted therapy than immunotherapy and should be treated with a TKI, even in the presence of high PD-L1 expression. Further evaluation of these agents in the neoadjuvant setting and combined with chemoradiotherapy is ongoing.

Cytotoxic Chemotherapy for Metastatic or Recurrent NSCLC

Cytotoxic chemotherapy is typically used in combination with immunotherapy as the initial treatment in patients with metastatic or recurrent NSCLC only when there is no contraindication to immunotherapy. Selected chemotherapy agents perform quite differently in squamous carcinomas versus adenocarcinomas. Patients with nonsquamous NSCLC have an improved survival when

TABLE 78-11 Results of Phase 3 Trials Comparing First-Line Immunotherapy with or without Chemotherapy Versus Chemotherapy Alone in Patients with NSCLC

STUDY	THERAPY	NO. OF PATIENTS	OS (MONTHS)	PFS (MONTHS)
Keynote 024	Pembrolizumab	154	30.0	7.9
PD-L1 ≥50%	Platinum-chemotherapy	151	14.2	3.5
Keynote 042	Pembrolizumab	637	16.7	5.4
PD-L1 ≥1%	Platinum-chemotherapy	637	12.1	6.5
IMPOWER 110	Atezolizumab	286	20.2	8.1
PD-L1 ≥50% TC or ≥15% IC	Platinum-chemotherapy	263	13.1	5.0
Keynote 189	Pembrolizumab + platinum-chemotherapy	410	NR	8.8
Nonsquamous	Platinum-chemotherapy	206	11.3	4.9
Keynote 407	Pembrolizumab + platinum-chemotherapy	278	15.9	6.4
Squamous	Platinum-chemotherapy	281	11.3	4.8
IMPOWER 150	Atezolizumab + platinum-chemotherapy	356	19.2	8.3
Nonsquamous	Platinum-chemotherapy	336	14.7	6.8
IMPOWER 130	Atezolizumab + platinum-chemotherapy	483	18.6	7.0
Nonsquamous	Platinum-chemotherapy	240	13.9	5.5
CheckMate 227	Nivolumab plus ipilimumab	583	17.1	5.1
	Platinum-chemotherapy	583	13.9	5.6
CheckMate-9LA	Nivolumab plus ipilimumab plus two cycles of platinum-chemotherapy	361	14.1	6.8
	Platinum-chemotherapy	358	10.7	5

Abbreviations: IC, immune cells; NR, not reported; OS, overall survival; PFS, progression-free survival; TC, tumor cells.

Note: Platinum-chemotherapy refers to first-line platinum doublet or triplet chemotherapy.

treated with cisplatin and pemetrexed compared to cisplatin and gemcitabine. By contrast, patients with squamous carcinoma have an improved survival when treated with cisplatin and gemcitabine. This survival difference is thought to be related to the differential expression between tumor types of thymidylate synthase (TS). Squamous cancers have a much higher expression of TS compared to adenocarcinomas, accounting for their lower responsiveness to pemetrexed. By contrast, the activity of gemcitabine is not impacted by the levels of TS.

Second-Line Therapy and Beyond Second-line therapy for advanced NSCLC relies on docetaxel; it improves survival compared to supportive care alone. Ramucirumab is a recombinant human IgG1 monoclonal antibody that targets VEGFR-2 and blocks the interaction of VEGF ligands and VEGFR-2. A phase 3 trial demonstrated a significant improvement in progression-free survival and overall survival when ramucirumab was combined with docetaxel as second-line therapy in patients who had progressed on platinum-based chemotherapy. Contrary to bevacizumab, ramucirumab was safe in patients with both squamous and nonsquamous NSCLC and is approved regardless of histology.

Supportive Care No discussion of the treatment strategies for patients with advanced lung cancer would be complete without a mention of supportive care. Coincident with advances in chemotherapy and targeted therapy was a pivotal study that demonstrated that the early integration of palliative care with standard treatment strategies improves both quality of life and overall survival for patients with stage IV NSCLC **(Chaps. 12** and **69)**. Aggressive pain and symptom control are important components of optimal treatment of these patients.

TREATMENT

Small-Cell Lung Cancer

The overall treatment approach to patients with SCLC is shown in **Fig. 78-9**.

SURGERY FOR LIMITED-DISEASE SMALL-CELL LUNG CANCER

SCLC is a highly aggressive disease characterized by its rapid doubling time, high growth fraction, early development of disseminated disease, and dramatic response to first-line chemotherapy and radiation. In general, surgical resection is *not* routinely recommended for patients because even patients with LD-SCLC still have occult micrometastases. However, the American College of Chest Physicians Evidence-Based Clinical Practice Guidelines recommend surgical resection over nonsurgical treatment in SCLC patients with clinical stage I disease after a thorough evaluation for distant metastases and invasive mediastinal stage evaluation (grade 2C). After resection, these patients should receive platinum-based adjuvant chemotherapy (grade 1C). If the histologic diagnosis of SCLC is made in patients on review of a resected surgical specimen, such patients should receive standard SCLC chemotherapy as well.

CHEMOTHERAPY

In patients with limited-stage SCLC, concurrent chemoradiotherapy with cisplatin-etoposide for four cycles has remained standard of care for over 4 decades. Two randomized phase 3 trials have demonstrated that chemotherapy with either cisplatin or carboplatin plus either etoposide and a PD-L1 inhibitor, atezolizumab (IMPOWER 133) or durvalumab (CASPIAN), provides superior progression-free and overall survival compared to chemotherapy alone, making combination therapy the preferred choice in appropriate patients. Despite response rates to first-line therapy as high as 80%, the median survival ranges from 12 to 20 months for patients with LD and approximately 12 months for patients with ED. Regardless of disease extent, the majority of patients relapse and develop chemotherapy-resistant disease. The prognosis is especially poor for patients who relapse within the first 3 months of therapy; these patients are said to have *chemotherapy-resistant disease*. Patients are said to have *sensitive disease* if they relapse >3 months after their initial therapy and are thought to have a somewhat better overall survival. These patients also are thought to have the greatest potential benefit from second-line chemotherapy. Topotecan and

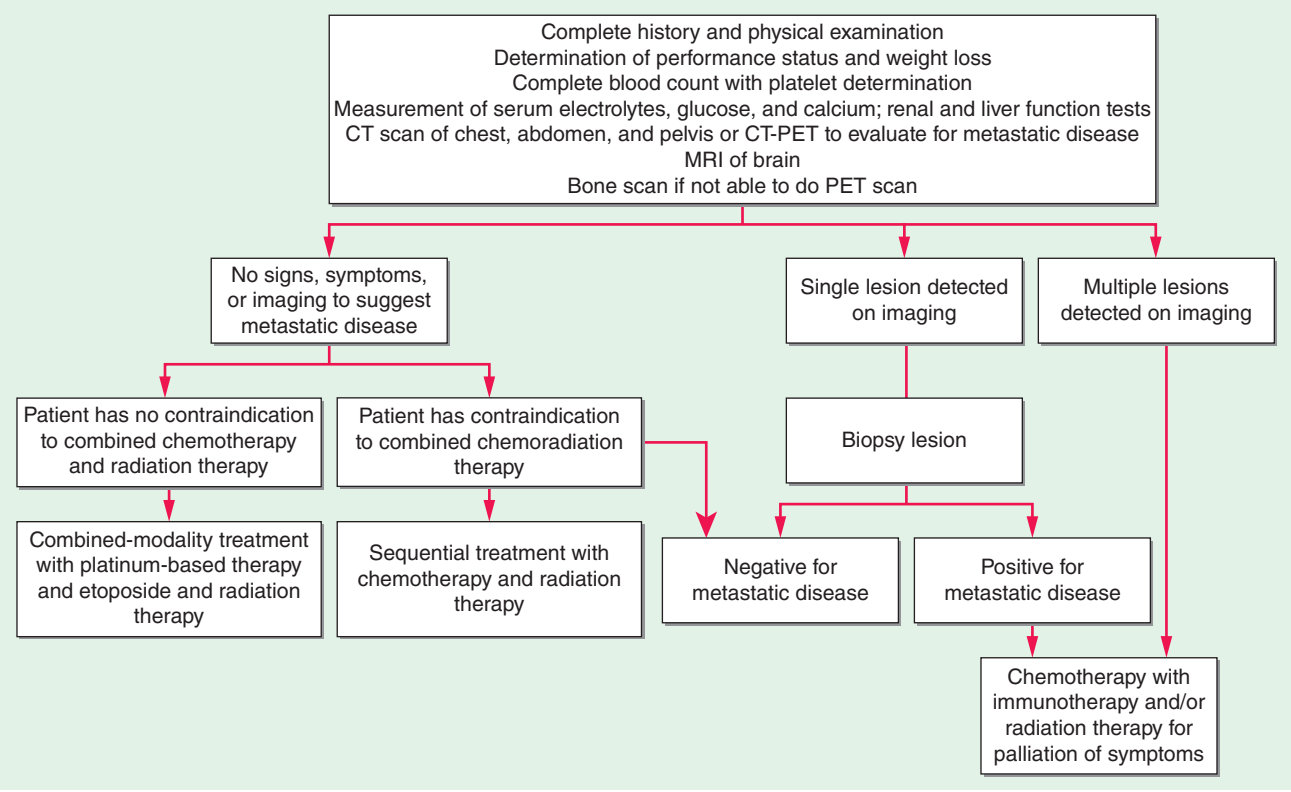


FIGURE 78-9 Algorithm for management of small-cell lung cancer. MRI, magnetic resonance imaging; PET, positron emission tomography.

lurbinectedin are FDA-approved agents for second-line therapy in patients with SCLC. Topotecan has only modest activity and can be given either intravenously or orally; it appears to have more efficacy in patients with chemotherapy-sensitive disease. Lurbinectedin has a 35% response rate and progression-free survival of 3.5 months, with a greater benefit in patients with chemotherapy-sensitive disease. Other agents with similar low levels of activity in the second-line setting include irinotecan, paclitaxel, docetaxel, vinorelbine, oral etoposide, and gemcitabine.

THORACIC RADIATION THERAPY

Thoracic radiation therapy (TRT) is a standard component of induction therapy for patients with good performance status and limited-stage SCLC. Meta-analyses indicate that chemotherapy combined with chest irradiation improves 3-year survival by ~5% as compared with chemotherapy alone. The 5-year survival rate, however, remains disappointingly low at ~10–15%. Most commonly, TRT is combined with cisplatin and etoposide chemotherapy due to a superior toxicity profile as compared to anthracycline-containing chemotherapy regimens. As observed in locally advanced NSCLC, *concurrent* chemoradiotherapy is more effective than *sequential* chemoradiation but is associated with significantly more esophagitis and hematologic toxicity. Ideally TRT should be administered with the first two cycles of chemotherapy because later application appears slightly less effective. If for reasons of fitness or availability, this regimen cannot be offered, TRT should follow induction chemotherapy. With respect to fractionation of TRT, twice-daily 1.5-Gy fractioned radiation therapy has been shown to improve survival in LD-SCLC patients but is associated with higher rates of grade 3 esophagitis and pulmonary toxicity. Although it is feasible to deliver once-daily radiation therapy doses up to 70 Gy concurrently with cisplatin-based chemotherapy, there are no data to support equivalency of this approach compared with the 45-Gy twice-daily radiotherapy dose. Therefore, the current standard regimen of a 45-Gy dose administered in 1.5-Gy fractions twice daily for 30 days is being compared with higher-dose regimens in two phase 3 trials, one in the United States and one in Europe. Patients should be carefully selected for concurrent chemoradiation therapy based on good performance status and adequate pulmonary reserve. The role of radiotherapy in ED-SCLC is largely restricted to palliation of tumor-related symptoms such as bone pain and bronchial obstruction.

PROPHYLACTIC CRANIAL IRRADIATION

Prophylactic cranial irradiation (PCI) should be considered in all patients either with LD-SCLC or who have responded well to initial therapy; its role in patients with ED-SCLC is more controversial. A meta-analysis including seven trials and 987 patients with LD-SCLC who had achieved a complete remission after upfront chemotherapy yielded a 5.4% improvement in overall survival for patients treated with PCI. In patients with ED-SCLC who have responded to first-line chemotherapy and had no CNS disease, patients randomized to observation had a higher incidence of brain metastases; however, use of PCI did not improve overall survival. Long-term toxicities, including deficits in cognition, have been reported after PCI but are difficult to sort out from the effects of chemotherapy or normal aging.

THYMIC TUMORS

Thymic tumors are rare malignancies accounting for 0.5–1.5% of all malignancies in the United States with a higher incidence among Asian populations. They are particularly rare among children and young adults with incidence peaking in the fifth decade of life. There is no difference between sexes, and no clear risk factors have been identified.

CLINICAL MANIFESTATIONS

The majority of thymic tumors occur in the anterior mediastinum. Approximately 40% of patients with mediastinal masses will be asymptomatic with an incidental finding on chest imaging. In patients presenting with an anterior mediastinal mass, if appropriate, serum β-human chorionic gonadotropin (HCG) and α fetoprotein (AFP) should be sent to rule out a germ cell tumor. A patient with a sign or symptom of thymoma or thymic carcinoma may present with chest pain, dyspnea, cough, or superior vena cava syndrome secondary to effects on adjacent organs or a paraneoplastic syndrome, most commonly myasthenia gravis, pure red cell aplasia, or hypogammaglobulinemia. More rare paraneoplastic syndromes include limbic encephalitis, aplastic anemia, hemolytic anemia, and autoimmune disease such as Sjögren's syndrome, polymyositis, rheumatoid arthritis, and ulcerative colitis, among others.

STAGING

Given the rarity of the tumor, patients with suspected thymoma should be evaluated by a multidisciplinary team including a surgeon, medical and radiation oncologist, and pathologist with experience in treating the disease. A CT scan of the chest with contrast is recommended to determine if the mass is resectable based on relationship to surrounding structures. An MRI with contrast may be performed if clinically indicated. A PET scan may be useful in the evaluation of a patient with thymic tumors, although it may be less useful in the staging of thymoma compared to thymic carcinoma. A core needle biopsy is considered standard of care for obtaining a histologic diagnosis of an anterior mediastinal tumor. This may be obtained via CT or ultrasound imaging. However, in some circumstances, a mediastinoscopy or open biopsy may be required.

Thymomas are commonly staged using the Masaoka system or the World Health Organization (WHO) staging system, as described in Table 78-12. WHO types A, AB, and B1 tend to be more well-differentiated, types B2 and B3 are moderately differentiated, and type C is poorly differentiated.

TREATMENT

Surgical resection is the mainstay of treatment for patients with Masaoka type I and II thymic tumors. In patients with type III and IV who have potentially resectable thymic tumors, neoadjuvant chemotherapy may be given to decrease the tumor size and allow for a resection with negative margins. Surgery remains controversial and provides a limited role in the treatment of stage III and IV disease. No

TABLE 78-12 Staging Thymic Tumors

MASAOKA STAGE	DEFINITION
I	Grossly and microscopically encapsulated
IIA	Microscopic transcapsular invasion
IIB	Macroscopic invasion into surrounding tissue excluding pericardium, lung, and great vessels
III	Macroscopic invasion into neighboring organs of the lower neck or upper chest
IVA	Pleural or pericardial dissemination
IVB	Hematogenous or lymphatic dissemination to distal organs
WHO	
A	Tumor with few lymphocytes
AB	Tumor with features of type A and foci rich in lymphocytes
B1	Tumor with features of normal epithelial cells with vesicular nuclei and distinct nucleoli and an abundant population of lymphocytes. Also known as cortical thymoma, lymphocyte-rich thymoma
B2	Thymoma with no or mild atypia with round or polygonal-shaped cells with small component of lymphocytes
B3	Well-differentiated thymic carcinoma with mild atypia
C	Thymic carcinoma with high atypia

additional therapy may be required in patients with type I who have a resection with negative margins. Postoperative radiation therapy may be recommended based on extracapsular extension and the presence of positive margins in patients with type II or III thymic tumors or histologic evaluation WHO B3 and C. Radiation therapy may be beneficial in patients with locally advanced disease (type III or IV) or in patients with symptoms secondary to compression of surrounding structures. Chemotherapy with cisplatin, doxorubicin, and cyclophosphamide (CAP) remains the mainstay of therapy in the neoadjuvant and adjuvant setting as well as first-line therapy in patients with metastatic thymoma, whereas carboplatin and paclitaxel are often employed in patients with thymic carcinoma. Limited additional agents are recommended based on small phase 2 trials as second-line therapy and beyond.

COVID-19 AND LUNG CANCER

COVID-19, a respiratory tract infection caused by SARS-CoV-2, emerged in Wuhan, China, in late 2019. The rapid global spread led the WHO to declare a pandemic in early March 2020. Large retrospective data sets have shown that cancer patients, and particularly patients with lung cancer, are at increased risk of morbidity and mortality from COVID-19. The dilemma of distinguishing COVID-19 symptoms from lung cancer and radiographic diagnosis of pneumonia or pneumonitis from radiation therapy or immunotherapy versus COVID-19 pneumonia has presented a particular challenge to health care providers. Mortality as high as 35% has been reported for patients with lung cancer infected with SARS-CoV-2. Older patients (≥65 years old), patients with a worse performance status (Eastern Cooperative Oncology Group performance status ≥1), patients on glucocorticoids (≥10 mg prednisone equivalent) and anticoagulation, and patients on chemotherapy within 3 months of diagnosis appear to be particularly at risk for morality if infected. The long-term impact on lung cancer mortality due to delays in screening, diagnosis, and treatment are likely to be felt for years to come.

SUMMARY

The management of SCLC and NSCLC has undergone major change in the past decade, resulting in a reduction in lung cancer mortality. For patients with early-stage disease, advances in radiotherapy and surgical procedures as well as new systemic therapies have greatly improved prognosis in all diseases. For patients with advanced lung cancer, major progress in understanding tumor genetics and tumor immunology has led to the development of rational targets and specific inhibitors, which have documented efficacy in specific subsets of NSCLC. Furthermore, increased understanding of how to activate the immune system to drive antitumor immunity has proven to be a successful therapeutic strategy for a subset of patients with advanced lung cancer. However, only a small subset of patients responds to immune checkpoint inhibitors, and the majority of patients treated with targeted therapies or chemotherapy eventually develop resistance, which provides strong motivation for further research and enrollment of patients onto clinical trials in this rapidly evolving area.

Acknowledgment

David Johnson and Christine Lovly contributed to this chapter in the prior edition, and material from that chapter has been retained here.

FURTHER READING

Drillon et al: Selpercatinib. Available at *https://www.nejm.org/doi/full/10.1056/NEJMoa2005653.*

Drilon et al: Entrectinib. Available at *https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(19)30690-4/fulltext*

Drilon et al: Larotrectinib. Available at *https://www.nejm.org/doi/full/10.1056/NEJMoa1714448.*

Ghandi L et al: Pembrolizumab plus chemotherapy in metastatic non-small cell lung cancer. N Engl J Med 378:2078, 2018.

Goldstraw et al: The IASLC Lung Cancer Staging Project: Proposals for Revision of the TNM Stage Groupings in the Forthcoming (Eighth) Edition of the TNM Classification for Lung Cancer. JTO 11:39, 2016.

Hellmann MD et al: Nivolumab plus ipilimumab in advanced non small cell lung cancer. N Engl J Med 381:2220, 2019.

Garasino et al: COVID. Available at *https://www.sciencedirect.com/science/article/pii/S1470204520303144?via%3Dihub.*

Horn L et al: First line atezolizumab plus chemotherapy in extensive stage small-cell lung cancer. N Engl J Med 379:2220, 2018.

Li et al: Trastuzumab-Deruxtecan. Available at *https://www.nejm.org/doi/full/10.1056/NEJMoa2005653.*

Paz-Ares L et al: Pembrolizumab plus chemotherapy for squamous non-small cell lung cancer. N Engl J Med 379:2040, 2018.

Paz-Ares L et al: Durvalumab plus platinum-etoposide versus platinume toposide in first line treatment of extensive-stage small-cell lung cancer (CASPIAN): A randomised, controlled, open-label, phase 3 trial. Lancet 394:1929, 2019.

Peters S et al: Alectinib versus crizotinib in untreated ALK-positive non-small cell lung cancer. N Engl J Med 377:829, 2017.

Reck M et al: Pembrolizumab versus chemotherapy for PD-L1 positive non-small cell lung cancer. N Engl J Med 375:1823, 2016.

Soria JC et al: Osimertinib in untreated EGFR-mutated advanced nonsmall cell lung cancer. N Engl J Med 378:113, 2018.

Wolf et al: Capmatinib. Available at *https://www.nejm.org/doi/full/10.1056/NEJMoa2005653.*

79 Breast Cancer

Daniel F. Hayes, Marc E. Lippman

INTRODUCTION AND BACKGROUND

CONCEPTUAL AND BIOLOGICAL ISSUES OF BREAST CANCER

Breast cancer is a malignant proliferation of epithelial cells lining the ducts or lobules of the breast. In the year 2020, approximately a quarter million cases of invasive and 61,000 cases of in situ breast cancer were diagnosed in the United States, with nearly 41,000 deaths. Epithelial malignancies of the breast are the most common cause of cancer in women (excluding skin cancer), accounting for about one-third of all cancer in women. As a result of earlier detection and improved treatments, the mortality rate from breast cancer decreased by more than one-third over the past three decades in high- and middle-income countries. This chapter does not consider rare malignancies presenting in the breast, such as sarcomas and lymphomas, but focuses on the epithelial cancers.

Breast cancer has served as a paradigm for several oncologic principles related to solid tumors. It spans a spectrum of conditions for which different clinical considerations must be made, including risk assessment, prevention, screening, evaluation of breast abnormalities, local and adjuvant systemic treatments, metastatic therapies, and survivorship issues **(Fig. 79-1)**.

The unique biology of breast cancer has rendered it amenable to a variety of therapeutic "targeted" strategies based on the appreciation of differences in subtypes that reflect the need for differences in assessment and therapy. These subtypes include expression of the estrogen receptor (ER) and the human epidermal growth factor receptor type 2 (HER2), as well as germline or somatic mutations in inherited tumor suppressor genes, such as *BRCA1* and *BRCA2*. Identifiable somatic mutations in genes that appear to drive the cancer, including mammalian target of rapamycin (*mTOR*), cyclin-dependent kinase 4 and 6 (*CDK4/6*), and S-phosphatidylinositol-4,5-bisphosphate 3-kinase